

Can open prescribing data detect medicine effects on tuberculosis? An ecological study of primary care and hospital prescribing and TB notifications in England

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Abstract

Background. - Corticosteroids, biologics and other immunosuppressants increase individual risk of tuberculosis (TB). - Openly published English prescribing and TB surveillance data could, in principle, be linked to study these relationships at population level. - We assessed whether such ecological analyses can detect associations between medicines and TB, and why they might not.

Methods. - Exposures. - Primary care prescribing, 2011–2024: practice-level data apportioned to local authorities by where registered patients live. Drug groups were defined at presentation level (systemic oral glucocorticoids, conventional DMARDs, transplant immunosuppressants and others), including a negative-control exposure. - Hospital medicines, 2019–2024: NHS Secondary Care Medicines Data in WHO defined daily doses, apportioned by trust catchment. Controls were a disease-specific positive control (active-TB treatment) and negative-control exposures. - **Outcomes.** Annual UKHSA TB notifications for 294 lower-tier and 151 upper-tier authorities, and by region, place of birth and age. - **Analysis.** - Poisson panel models with area and year fixed effects, exposure in the previous year, and falsification (lead) tests. - Regional models with randomisation inference. - Minimum detectable effects (MDE), compared with expected population effects derived from published relative risks and UKHSA-recorded drug-associated TB. - A permutation-based power simulation on the real notification counts.

Results. - Linkage. Hospital active-TB treatment tracked TB notifications between areas (Spearman $\rho = 0.79$) and within areas in the same year (incidence rate ratio [IRR] 1.035, 95% CI 1.013–1.058, per 10% increase). The within-area elasticity was only 0.36–0.40, indicating substantial attenuation even for a disease-specific drug. - **Primary care.** No drug group was associated with notifications in the following year. For systemic oral glucocorticoids the IRR was 0.982 (0.941–1.025) per 10% within-area increase. - **Hospital medicines.** TNF inhibitors (0.989, 0.972–1.006), JAK inhibitors and systemic glucocorticoids were also null. - **Falsification and negative control.** Several falsification tests failed, and the negative-control exposure showed inverse estimates similar in size to candidate drugs. This indicates residual confounding by area-specific trends. - **Power.** - For oral glucocorticoids, the MDE (6.3% per 10% increase) was about 18 times the effect expected from published relative risks (0.34%), and about 110 times that implied by UKHSA-recorded steroid-associated TB (0.055%). - In simulations on the real counts, the published effect was detected in 7.7% of replicates, no better than the 7.0% false-positive rate with no effect. - **UK-born TB.** A regional association between prednisolone dose and UK-born TB was implausibly large and was accompanied by failed negative-control and falsification tests.

Conclusions. - Open English prescribing and TB notification data can be linked, but only with heavy attenuation, and ecological analyses of them cannot detect plausible population-level effects of medicines on TB. - Apparent associations reflect confounding by migration, age and area trends. - Causal questions require individual-level linked data analysed with target trial

emulation. Surveillance of drug-associated TB is better served by more complete recording of immunosuppression in national TB surveillance.

Introduction

Tuberculosis (TB) remains a public health problem in England.

Recent trends. - **Decline:** notifications fell by almost half between 2011 and 2018. The fall was largely explained by fewer recent migrants from high-incidence countries and by pre-entry screening, extended in 2012 to long-term visa applicants from high-incidence countries [Thomas 2018; Aldridge 2016]. - **2020:** notifications dropped to 4,125 during the COVID-19 pandemic, a fall thought to reflect service disruption rather than less disease [UKHSA 2021; Morrison 2023]. - **2021-2024 rise:** notifications then increased, reaching 5,490 in 2024 (9.4 per 100,000). - 81.5% were in people born outside the UK. - 41% of those were notified within five years of arrival. - The rise has been attributed mainly to migration from higher-incidence countries [UKHSA 2025]. - **2025:** national data show 5,424 notifications, with 2024 revised to 5,487 [UKHSA 2026].

TB in UK-born people. UK-born notifications fell by 43% between 2014 and 2022, then rose [UKHSA 2025]. UK-born TB is heterogeneous: it includes reactivation in older adults, transmission linked to social risk factors, and disease in UK-born children of migrant families [Davidson 2018].

Medicines that change TB risk. Host factors also modify TB risk, and several are shaped by commonly prescribed medicines. - **Oral glucocorticoids:** current use is associated with an odds ratio of about 5 for active TB, higher at doses ≥ 15 mg/day [Jick 2006]. - **Inhaled corticosteroids and conventional DMARDs:** smaller increases [Brassard 2011; Castellana 2019; Brassard 2006]. - **Anti-TNF biologics:** TB typically presents within months of starting treatment [Keane 2001]. Latent infection screening before treatment greatly reduces this risk [BTS 2005; Carmona 2005]. - **Proton pump inhibitors:** associations with increased risk [Song 2019]. - **Statins and metformin:** associations with lower risk [Li 2020; Zhang 2020].

Many of these estimates come from high-incidence settings or designs vulnerable to confounding by indication and protopathic bias.

Open data and their problems. England publishes monthly prescribing for every general practice, hospital medicines use by NHS trust, and TB notifications by local authority. Linking these offers a low-cost way to generate or test hypotheses about medicines and TB. Such ecological analyses face well-known problems: - confounding by area characteristics; - cross-level bias, which covariate adjustment cannot remove when effects differ between groups [Greenland 1989; Morgenstern 1995]; - limited power, which depends on the number of areas [Sheppard 1996].

The data carry specific problems too: - practice lists are inflated in ways that track population mobility [Hudson 2025; Venkatesan 2025]; - practice locations do not match where patients live; - prescribing is patterned by deprivation and was disrupted by COVID-19 [Richards 2025]; - hospital-prescribed medicines are recorded separately from primary care [Bacon 2020].

The gap. We found no previous study linking area-level prescribing to TB. In analogous settings, drugs with large individual effects produced weak population signals [Hermans 2015], and between-area and within-area associations had opposite signs [Költringer 2023].

This study. We asked whether open English prescribing data can detect associations between medicines and TB notifications at population level, and why they might not. We combined: - cross-sectional and fixed-effects panel designs at two geographic scales; - primary care prescribing apportioned to where registered patients live; - hospital medicines apportioned by trust catchment, with a disease-specific positive control; - negative-control exposures and falsification tests [Lipsitch 2010; Prasad 2013]; - regional analyses of TB in UK-born and older people.

We compared the smallest effects these designs could detect, both analytically and by simulation,

with the population effects implied by published individual-level relative risks and by UKHSA surveillance records of drug-associated TB.

Methods

Design and reporting

This study uses aggregate, openly published data in three linked ecological designs:

- a cross-sectional analysis;
- a longitudinal panel analysis of local authorities with area and year fixed effects;
- regional analyses by place of birth and age.

It was reported with reference to the RECORD and STROBE guidance. All data were public, aggregate and anonymised, so ethical approval was not required. Code, processed data and outputs are available at <https://github.com/drcjar/tb-prescribing-england>.

Pre-specification. The analysis developed in stages, and we report the chronology.

1. **First pass.** Before any results were seen, we specified the drug groups (including a negative-control exposure), the cross-sectional design and the first panel design. The first panel design used three-year rolling notifications and exposure in years $t-5$ to $t-3$.
2. **Later additions, made after those analyses gave null results:**
 - annual outcomes;
 - lower-tier authorities;
 - hospital medicines and their controls;
 - regional analyses by birthplace and age;
 - power simulation;
 - spatial diagnostics.
3. **Revisions after peer review:**
 - drug groups redefined;
 - prescribing apportioned by patient residence;
 - exposure in year $t-1$ as primary;
 - lag and lead estimated jointly;
 - randomisation inference;
 - additional covariates.

All analyses after the first pass are exploratory. Estimates from the original designs are given in the supplement.

Assumed causal structure

Figure 1 sets out the assumed causal structure. It shows which factors are absorbed by area and year fixed effects, which are measured and adjusted for, and which remain unmeasured. The primary exposure is prescribing in the year before the outcome year, because same-year prescribing may be prescribing for undiagnosed TB (protopathic bias).

TB notifications

TB notifications are cases reported to UKHSA surveillance: the Enhanced TB Surveillance system (ETS) until 2021, and the National TB Surveillance System (NTBS) since, into which data from 2018 onwards were migrated. Cases are culture-confirmed or clinically diagnosed, and are assigned to areas by residential postcode at notification.

We used four sources:

1. **Annual counts by local authority, 2001-2024.**
 - *Source:* UKHSA TB regional reports 2024, supplementary tables.

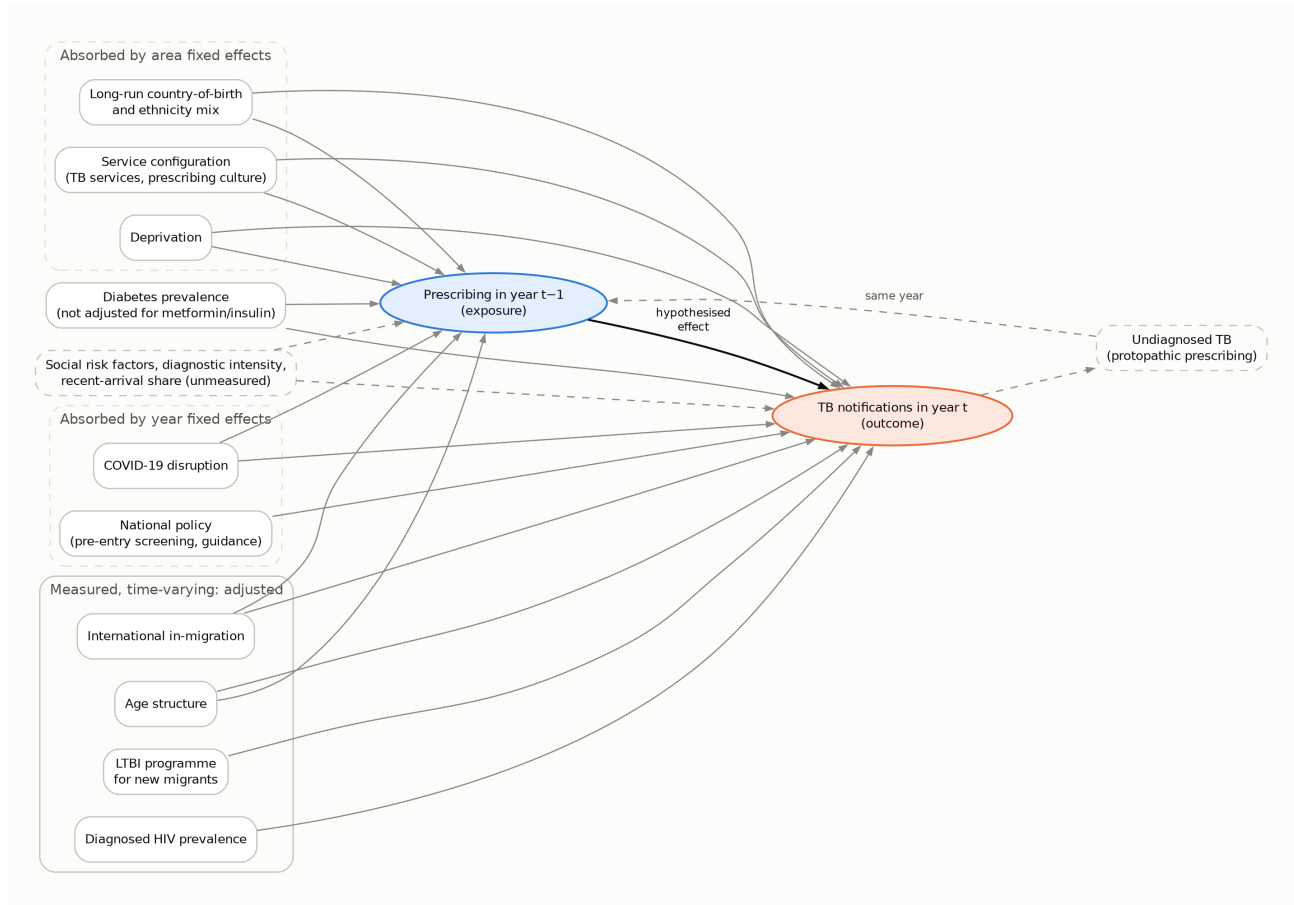


Figure 1: **Figure 1.** Assumed causal structure for area-level prescribing and TB notifications. Area fixed effects absorb stable area characteristics; year fixed effects absorb national shocks; time-varying measured factors are adjusted for; dashed nodes are unmeasured. Same-year pre-prescribing for undiagnosed TB (protopathic bias) motivates the use of prescribing in the previous year.

- *Coverage*: upper-tier authorities (UTLAs; 151, of which 149 were analysed) and lower-tier authorities (LTLAs; 294, of which 292 were analysed).
 - *Boundary handling*: counts are published on April 2023 boundaries. The City of London is combined with Hackney, and the Isles of Scilly with Cornwall.
 - *Consistency*: summed over three years, these counts matched the Fingertips three-year counts ($r = 0.996$). Local authority sums slightly exceed national totals (5,539 vs 5,490 in 2024), reflecting different extract dates.
2. **Region × place of birth × year, 2000-2024**, with Labour Force Survey denominators: *TB in England 2025*, Supplementary Table 12.
 3. **Region × place of birth × age group × year**: regional reports, Table 9.
 4. **National notifications with recorded immunosuppression by cause, 2024**: *TB in England 2025*, chapter 1.

The 2020 fall in notifications is thought to reflect service disruption rather than a true reduction in incidence.

Primary care prescribing

Sources. - **2010-2013**: HSCIC practice-level prescribing (PDPI), August 2010–December 2013.
- **2014-2024**: NHSBSA English Prescribing Dataset (EPD).

Both record items dispensed in the community from primary care prescriptions. We aggregated each month to practice level using identical drug-group rules. For the EPD this was done with server-side SQL.

Practices included. We restricted to standard GP practices (NHS ODS prescribing setting R076), which accounted for 98–99% of items after 2014.

Drug groups. Groups were defined at BNF presentation level:

Group	Definition
Systemic oral glucocorticoids	Prednisolone, prednisone, methylprednisolone, deflazacort and dexamethasone. Injectables, hydrocortisone (mainly replacement therapy) and betamethasone soluble tablets excluded. Oral hydrocortisone and oral dexamethasone are also reported separately.
Inhaled corticosteroids	—
Conventional DMARDs	Rheumatology methotrexate (including subcutaneous), leflunomide, azathioprine, mercaptopurine.
Transplant immunosuppressants	Tacrolimus, ciclosporin, mycophenolate, sirolimus, everolimus.
Other groups	Proton pump inhibitors; statins; metformin; insulins; fluoroquinolones; all antibacterials; vitamin D.
Descriptive only	Antituberculosis drugs: TB treatment in England is delivered by specialist services, so primary care prescribing of these drugs is not a valid positive control.
Negative control exposure	Levothyroxine.

Eye, ear, nose and skin preparations were excluded.

Exposure measures. - Items per 1,000 residents per year (primary). - Prednisolone-equivalent milligrams for systemic oral glucocorticoids: strength × quantity × potency, with dexamethasone 6.67, methylprednisolone 1.25 and deflazacort 0.83. - Average daily quantities (ADQ) for the groups where the EPD populates them.

Apportionment to areas. Each practice's annual prescribing was shared across local authority districts in proportion to where its registered patients lived, using NHS Digital counts of patients registered at each practice by LSOA: - April releases from 2014 to 2024 were used; - 2014 shares were applied to 2011–2013; - LSOAs were mapped to April 2023 districts.

Assigning each practice to the district containing its postcode was a sensitivity analysis. In 2024, 7.3% of registered patients lived outside the district of their practice.

Denominators. ONS mid-year resident population estimates.

Hospital medicines

Source and period. NHS Secondary Care Medicines Data (SCMD; NHSBSA), which gives trust × month × product quantities. We used January 2019 to December 2024.

Classification. Products were grouped by mechanism, with patient-year equivalents calculated from WHO defined daily doses (DDD), or documented maintenance doses where WHO gives none. The groups were:

- TNF inhibitors;
- IL-6 inhibitors and abatacept;
- JAK inhibitors used in rheumatology (ruxolitinib reported separately);
- rituximab;
- calcineurin and mTOR inhibitors;
- antiproliferatives;
- systemic glucocorticoids, measured in prednisolone-equivalent mg and excluding intra-articular, depot and topical forms.

Controls. - **Positive control:** active-TB treatment, defined as pyrazinamide- or ethambutol-containing products, including the fixed-dose combinations Rifater and Voractiv. - **Reported separately:** rifamycin and isoniazid products, which are also used for latent TB and, in the case of rifampicin, other infections. - **Negative-control exposures:** IL-17, IL-12/23 and α4β7 inhibitors and anakinra (low TB risk), and levetiracetam.

Trust mergers. Trusts that merged during 2019–2024 were reassigned to their successors using NHS ODS records. With this, 98–100% of DDD in every group and year came from trusts with catchment data.

Apportionment to areas. Trust quantities were shared across local authorities using OHID acute trust catchment populations (2024, all admissions). Elective-admission catchments were a sensitivity analysis.

Covariates

Time-varying (panel analyses): - ONS mid-year population estimates by age (% aged ≥65; % aged 15–44); - international in-migration per 1,000 (ONS components of population change); - diagnosed HIV prevalence; - QOF diabetes prevalence; - the latent TB infection (LTBI) testing and treatment programme for new migrants, as an active-area indicator mapped from CCGs to local authorities (active in 83 LTLAs); - people receiving asylum support per 1,000 (Home Office; from 2014).

Cross-sectional analyses: - Census 2021 percentage born outside the UK.

Statistical analysis

Panel analyses. - **Model:** Poisson pseudo-maximum-likelihood regression of annual notifications, with area and year fixed effects, a log population offset and SEs clustered by area. - **Primary exposure:** log prescribing rate in year $t-1$. Drug-associated TB typically presents within months of exposure [Keane 2001], and same-year prescribing is affected by prescribing for undiagnosed TB (protopathic bias). - **Outcome years:** 2014–2024. - **Covariates:** the time-varying set above. Diabetes prevalence was not adjusted for in the metformin and insulin models. - **Effect measure:** incidence rate ratio (IRR) per 10% within-area increase in prescribing, with Benjamini-Hochberg false discovery rate (FDR) correction across candidate drug groups. Controls were excluded from the FDR correction. - **Falsification:** prescribing in year $t-1$ and year $t+1$ estimated jointly on a common sample; a true effect should load on $t-1$ only. - **Sensitivity analyses:** - no covariates; - a distributed lag (t , $t-1$, $t-2$); - the original three-year window ($t-3$ to $t-1$); - area-specific linear trends; - log population as a covariate instead of an offset (shared-denominator bias); - adjustment for the LTBI programme; - adjustment for asylum support; - excluding outcome years 2020–21 and exposure years 2020–21; - outcome years 2018–2024 only (NTBS-era data); - outcome years 2015–2024 only, so that all exposure comes from the EPD (splice sensitivity); - ADQ instead of items, for groups where the EPD populates ADQ; - practice-postcode apportionment; - UTLA geography. - **Spatial dependence:** Moran's I of year-specific Pearson residuals (5-nearest-neighbour weights, 999 permutations).

Hospital analyses. The same model was fitted for 2019–2024, with exposure in: - the same year (for the positive control, where treatment follows diagnosis); - the previous year (for candidate drugs); - the following year.

SEs were also clustered by each area's principal trust, because apportioned exposures share trust-level variation. The within-area elasticity of the positive control (log IRR per log unit of exposure) was used as an attenuation factor.

Regional analyses. - **Units:** 9 regions $\times$ year. - **Outcomes:** all TB notifications; UK-born and non-UK-born notifications; UK-born notifications at age ≥ 65 (denominator: all residents aged ≥ 65). - **Model:** Poisson regression with region and year fixed effects, exposure lagged or led 1–3 years, adjusted for in-migration and age structure. - **Inference:** with only nine clusters, cluster-robust $t(8)$ intervals and randomisation p-values, permuting whole regional exposure histories across regions (499 permutations). - **Falsification:** lag 2 and lead 2 estimated jointly.

Cross-sectional analyses. Negative binomial regression of TB notifications on standardised log prescribing rates for the same period, adjusted for percentage born outside the UK, age structure, in-migration, HIV and diabetes prevalence.

Expected effects and minimum detectable effects (MDEs).

Expected effects. The population effect of a 10% increase in use was calculated as $[1 + 1.1p(RR - 1)] / [1 + p(RR - 1)]$, where p is prevalence of use and RR the individual-level relative risk (odds ratios where only these were available). The calculation assumes that:

1. prevalence of use scales with the prescribing measure;
2. baseline risk is homogeneous;
3. timing is aligned.

All three favour detection. Alternative expected effects came from three sources:

- recorded drug-associated notifications;
- stratification by age and place of birth;
- SCMD-derived prevalence of hospital drug use, including screening-attenuated relative risks for TNF inhibitors and attenuation by the positive-control elasticity.

MDEs. The MDE was calculated as $\exp(2.80 \times SE)$. For each MDE we report:

- the minimum detectable population attributable fraction (PAF);
- the largest PAF compatible with the upper 90% confidence limit.

Power simulation. A permutation-based simulation kept the real TB counts, so real overdispersion, serial correlation and trends were preserved. Whole exposure trajectories were permuted across areas, and known effects were injected by adding cases. This gave 500 null and 300 effect replicates per scenario, including scenarios where the effect acts through same-year exposure.

Analyses used Python 3.14 (pandas 3.0, statsmodels).

Results

Prescribing, TB notifications and within-area variation

TB notifications in England - Fell from 6,474 in 2014 to a low of 4,125 in 2020. - Rose to 5,490 in 2024, when 81.5% of people notified were born outside the UK.

Within-area variation in prescribing - Once prescribing was apportioned by where patients live, it varied little within areas over time. - For systemic oral glucocorticoids, the SD of the log rate after removing area and year means was 0.043, against 0.29 between areas. 95% of area-years were within $\pm 9\%$ of their expected value. - PPIs (0.038) and levothyroxine (0.043) were similar; conventional DMARDs varied more (0.089). - A 10% within-area change in prescribing is therefore at the edge of the observed data (Figure 2).

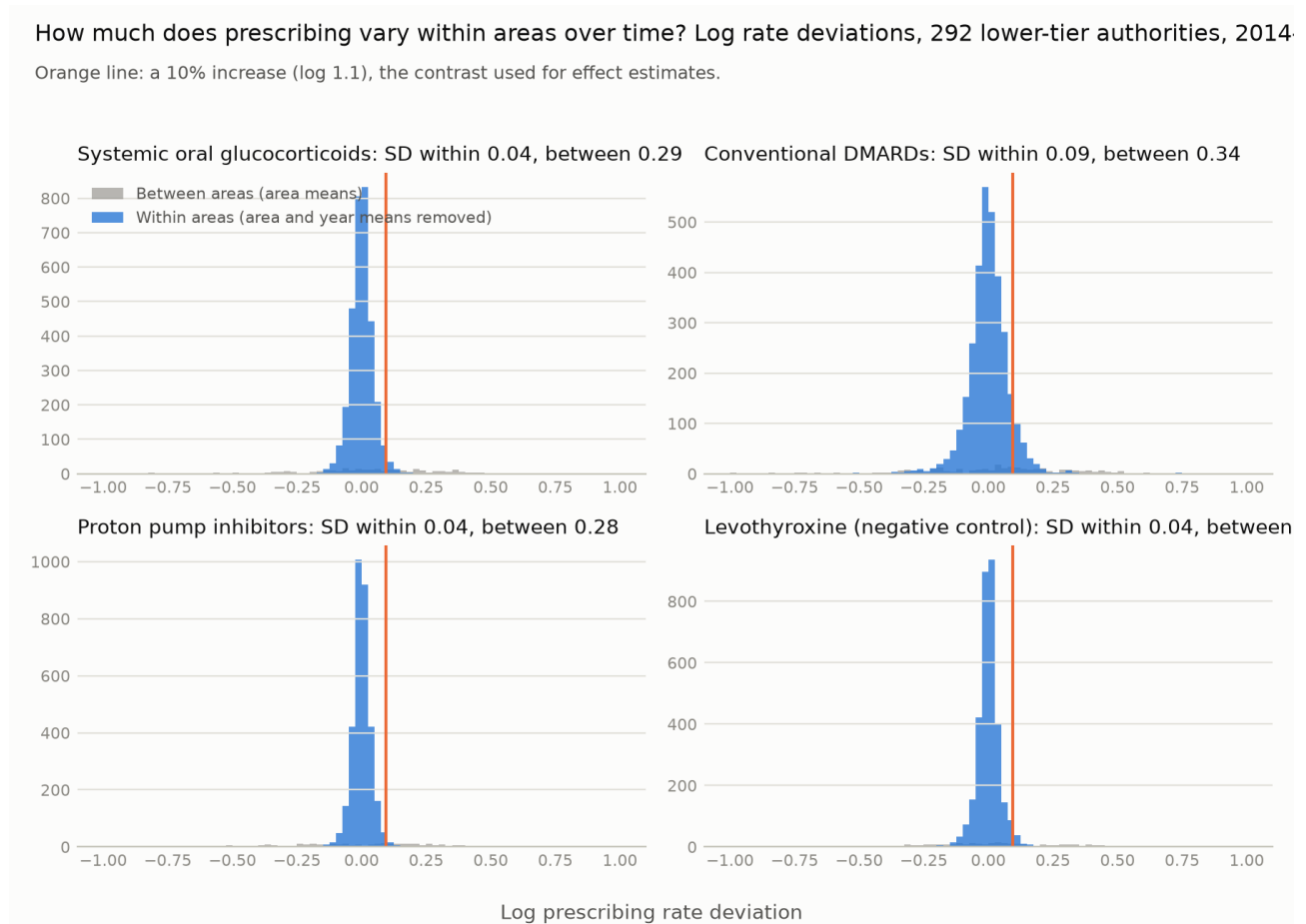


Figure 2: **Figure 2.** Within-area versus between-area variation in log prescribing rates, 292 lower-tier authorities, 2014–2024. Orange line: a 10% increase.

Primary care prescribing and TB notifications (panel analyses)

Primary analysis (294 lower-tier authorities, 3,233 area-years, outcome years 2014–2024) - No drug group was associated with notifications in the following year (Table 1; all FDR $q \geq 0.77$). - Systemic oral glucocorticoids: IRR 0.982 (95% CI 0.941–1.025) per 10% within-area increase.

Other groups (same scale, 95% CI):

Drug group	IRR
Oral hydrocortisone	1.002 (0.990–1.014)
Inhaled corticosteroids	0.981 (0.935–1.029)
Conventional DMARDs	1.003 (0.984–1.023)
Transplant immunosuppressants	0.999 (0.993–1.005)
Proton pump inhibitors	1.009 (0.967–1.054)
Statins	1.000 (0.960–1.041)
Metformin	0.969 (0.938–1.001)
Levothyroxine (negative control)	0.970 (0.938–1.003)

- Metformin’s inverse point estimate was matched in size by the negative control, levothyroxine.

Table 1. Primary care prescribing in the previous year and TB notifications: incidence rate ratio per 10% within-area increase (95% CI)

Drug group	LTLA, residence (294 areas, 2014–2024)	UTLA, residence	LTLA, practice postcode	Falsification: following year (t+1, joint with t–1)	Largest compatible PAF (90% upper limit)
Antituberculosis (descriptive)	1.000 (0.997–1.003)	1.000 (0.997–1.004)	1.001 (0.998–1.004)	1.001 (0.998–1.005)	3%
Systemic oral glucocorti- coids	0.982 (0.941–1.025)	0.987 (0.940–1.037)	0.983 (0.948–1.019)	0.946 (0.906–0.987)	18%
Oral hydro- cortisone	1.002 (0.990–1.014)	0.998 (0.983–1.014)	1.002 (0.991–1.014)	1.006 (0.990–1.022)	12%
Oral dexam- ethasone	1.004 (0.994–1.015)	1.005 (0.993–1.018)	1.002 (0.993–1.012)	1.003 (0.993–1.014)	13%
Inhaled corti- costeroids	0.981 (0.935–1.029)	0.976 (0.927–1.028)	0.983 (0.944–1.024)	0.875 (0.837–0.914)	21%
Conventional DMARDs	1.003 (0.984–1.023)	1.003 (0.980–1.026)	1.005 (0.988–1.023)	0.984 (0.950–1.019)	19%
Transplant immunosup- pressants	0.999 (0.993–1.005)	0.999 (0.992–1.005)	1.000 (0.995–1.006)	0.999 (0.991–1.006)	4%
Proton pump inhibitors	1.009 (0.967–1.054)	1.004 (0.960–1.050)	1.001 (0.963–1.040)	0.969 (0.908–1.034)	46%
Statins	1.000 (0.960–1.041)	0.997 (0.955–1.042)	0.994 (0.959–1.031)	0.992 (0.929–1.060)	34%
Metformin	0.969 (0.938–1.001)	0.964 (0.929–1.000)	0.967 (0.938–0.996)	0.998 (0.945–1.054)	0%
Insulins	0.981 (0.945–1.019)	0.987 (0.945–1.030)	0.983 (0.952–1.015)	0.903 (0.852–0.957)	13%
Fluoroquinolones	1.010 (0.997–1.023)	1.007 (0.993–1.021)	1.008 (0.995–1.020)	0.994 (0.980–1.009)	20%

Drug group	LTLA, residence (294 areas, 2014-2024)	UTLA, residence	LTLA, practice postcode	Falsification: following year (t+1, joint with t-1)	Largest compatible PAF (90% upper limit)
All	1.012	1.016	1.004	0.993	47%
antibacterials	(0.971-1.054)	(0.970-1.064)	(0.967-1.042)	(0.962-1.025)	
Vitamin D	1.008	1.011	1.007	1.004	25%
	(0.989-1.029)	(0.989-1.033)	(0.989-1.026)	(0.978-1.031)	
Levothyroxine	0.970	0.961	0.971	0.967	0%
(negative control)	(0.938-1.003)	(0.930-0.994)	(0.942-1.000)	(0.909-1.030)	

Poisson PML with area and year fixed effects, adjusted for age structure, international immigration, HIV and diabetes prevalence (diabetes not adjusted for metformin and insulins); SEs clustered by area. PAF, population attributable fraction implied by the upper 90% confidence limit.

Largest attributable fraction compatible with the data (upper 90% confidence limit) - Systemic oral glucocorticoids: 18% - Conventional DMARDs: 19% - Transplant immunosuppressants: 4%

Robustness - Upper-tier geography (151 areas, 1,661 area-years) gave similar results: - systemic oral glucocorticoids 0.987 (0.940-1.037); all FDR $q \geq 0.62$ - the negative control, levothyroxine, was nominally inverse at 0.961 (0.930-0.994), the same size as metformin (0.964, 0.929-1.000). This suggests residual confounding by trends. - Apportioning prescribing by practice postcode gave similar results: lower-tier oral glucocorticoids 0.983 (0.948-1.019). - Estimates were also materially unchanged by: - a three-year prior exposure window - adjustment for the LTBI programme or asylum support - excluding COVID-affected years - log population as a covariate - restricting to outcome years 2018-2024 (oral glucocorticoids 0.986, 0.944-1.030) - restricting to EPD-era exposure (0.991, 0.950-1.033) - ADQ instead of items (Table S1) - Area-specific linear trends moved inhaled corticosteroids to 1.053 (1.003-1.105).

Falsification tests (prescribing in years t-1 and t+1 estimated jointly) - Several drug groups showed opposite-signed lag and lead estimates, a pattern expected from shared trends rather than drug effects: - inhaled corticosteroids: t-1 1.076 (1.017-1.138), t+1 0.875 (0.837-0.914) - insulins: t-1 1.055 (1.006-1.106), t+1 0.903 (0.852-0.957) - systemic oral glucocorticoids: t+1 0.946 (0.906-0.987) - Across all 195 estimates in the LTLA residence models, 11 were nominally significant, most of them lead or distributed-lag terms.

Spatial dependence - Raw TB notification rates were strongly spatially clustered in every year (Moran's I 0.25-0.39 lower-tier; 0.40-0.49 upper-tier). - Residuals from the primary model were not: - lower-tier: median I 0.028 (range -0.016 to 0.167), nominally significant in 3 of 11 years - upper-tier: median I 0.017, nominally significant in 1 of 11 years.

Oral glucocorticoid prescribing trends and changes in TB notifications

National trends, 2011-2024 (standard GP practices) - Systemic oral glucocorticoid items rose from 115 to 132 per 1,000 residents in 2011-2016, then fell to 111 in 2024. Most of the fall came in 2020-21 (Figure 3). - Prednisolone-equivalent mg per item fell from 225 to 192, so courses became smaller. - Oral hydrocortisone items, mainly replacement therapy, rose from 5.3 to 9.1 per 1,000.

Correlation with national TB trends - National notification rates and prescribing were not correlated in levels (14 years; $r = 0.17$). - Year-on-year changes were not correlated at lags of

TB notifications and primary care prescribing: lower-tier local authorities, England

Annual notifications 2014-2024, 294 areas, up to 3,234 area-years; prescribing apportioned by residence. Poisson PML with area and year fixed effects; adjusted for age structure, international in-migration, HIV and diabetes prevalence (not diabetes for metformin and insulins); SEs clustered by area.

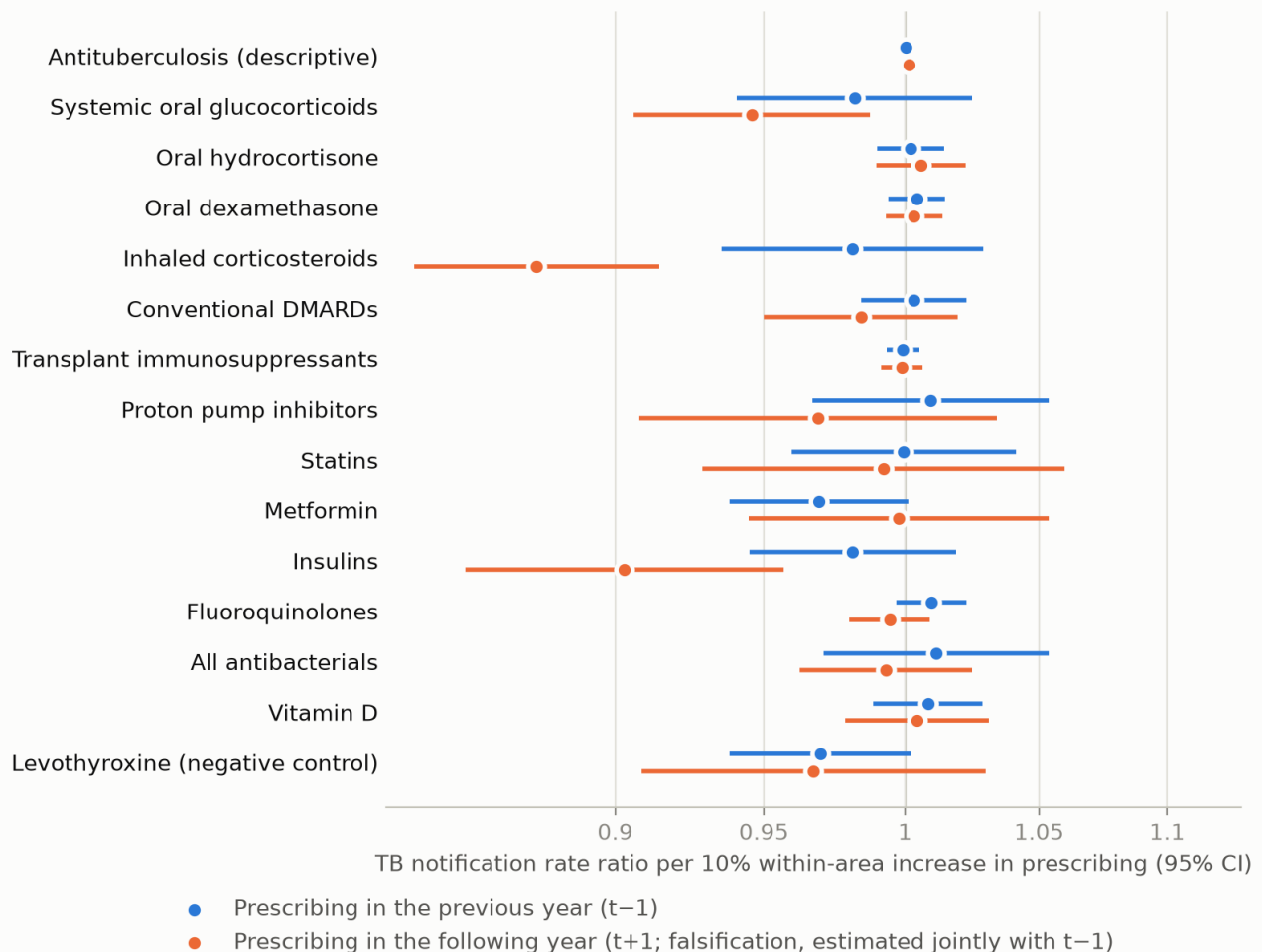


Figure 3: **Figure 4.** Primary care prescribing and TB notifications, lower-tier authorities, prescribing apportioned by patient residence. Blue: prescribing in the previous year (primary). Orange: prescribing in the following year, estimated jointly (falsification).

Oral glucocorticoid prescribing in English primary care, 2011–2024

Standard GP practices; HSCIC practice-level prescribing (2011–13) and NHSBSA English Prescribing Dataset (2014–24); per resident using ONS mid-year estimates. Systemic = prednisolone, prednisone, methylprednisolone, deflazacort, dexamethasone.

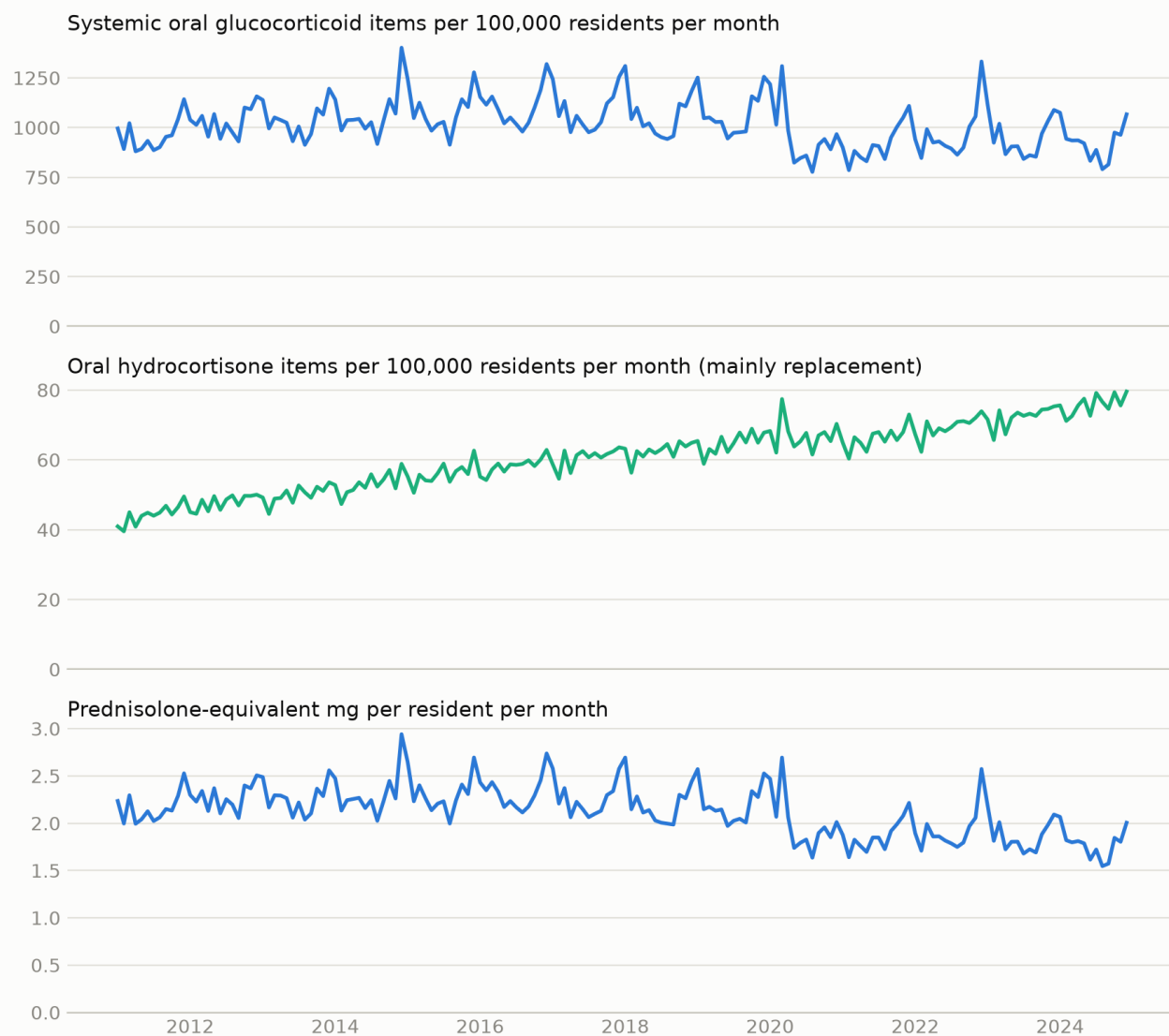


Figure 4: **Figure 3.** Oral glucocorticoid prescribing in English primary care, 2011–2024: systemic oral glucocorticoid items, oral hydrocortisone items and prednisolone-equivalent mg per resident.

0–2 years. - At a 3-year lag, one of eight correlations was nominally significant and inverse ($r = -0.75$).

Regional and local changes - In the regional panel (9 regions, 2012–2024), adjusted estimates were null: - items, lag 1: 1.008 (0.764–1.331); randomisation $p = 0.95$ - prednisolone mg, lag 1: 1.064 (0.843–1.344). - Across 140 upper-tier authorities, changes in prescribing (2014–16 to 2019–21) and in notifications (2014–16 to 2022–24) were weakly inversely related before adjustment (Spearman $\rho = -0.17$). After adjustment for change in in-migration they were unrelated (ratio per 10% 0.989, 0.925–1.057).

Hospital medicines

Coverage - After reassigning merged trusts to their successors, 98–100% of defined daily doses (DDD) in every drug group and year came from trusts with catchment data (Table S7).

Positive control: active-TB treatment (upper-tier authorities) - Between areas: tracked notifications (Spearman $\rho = 0.79$; adjusted IRR per SD 1.17, 1.03–1.32). - Within areas, same year: 1.035 (1.013–1.058) per 10% increase. - Previous year: 1.005 (0.995–1.014). Following year, estimated jointly: 1.000 (0.988–1.012). - The within-area elasticity was 0.36 (0.40 at lower-tier level): even a disease-specific drug was recovered with roughly 60% attenuation. - Rifamycin/isoniazid products behaved similarly (same year 1.035).

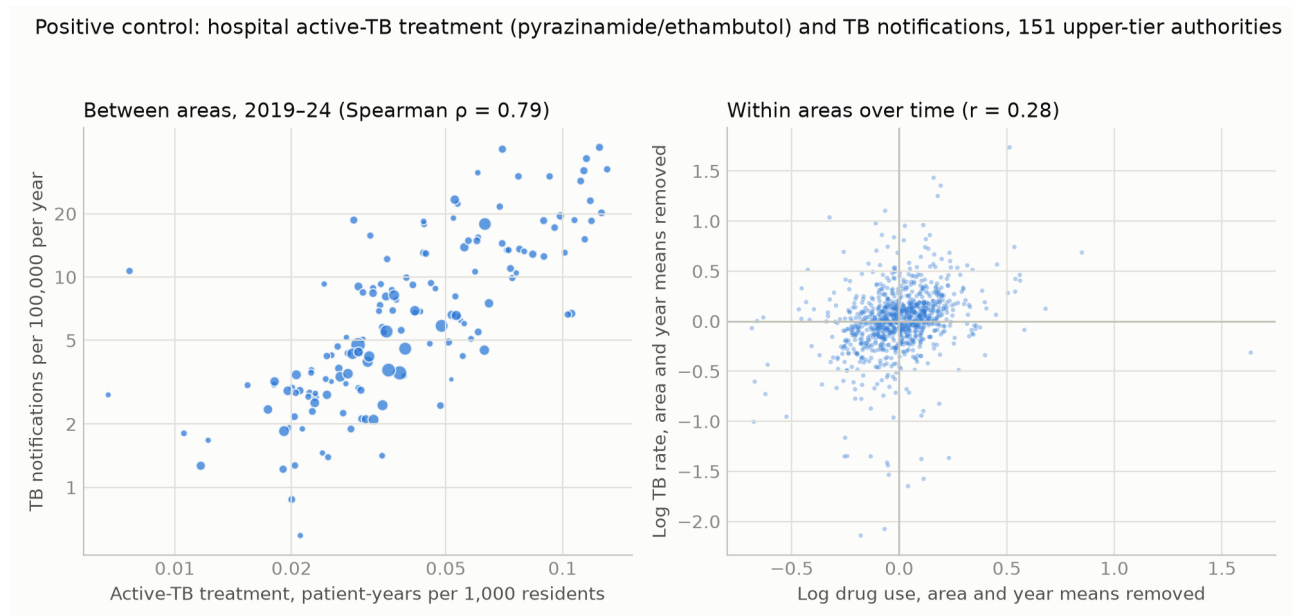


Figure 5: **Figure 5.** Positive control: hospital active-TB treatment (pyrazinamide/ethambutol-containing products, patient-year equivalents per 1,000 residents) and TB notifications between areas (left) and within areas over time (right).

Candidate drugs - No drug group with plausible TB risk was associated with notifications in the following year (Table 3). At upper-tier level, per 10% increase:

Drug group	IRR, previous year
TNF inhibitors	0.989 (0.972–1.006)
IL-6 inhibitors/abatacept	1.000 (0.992–1.008)
JAK inhibitors	0.994 (0.987–1.001)
Rituximab	0.998 (0.985–1.011)
Calcineurin/mTOR inhibitors	1.008 (0.996–1.020)

Drug group	IRR, previous year
Antiproliferatives	0.996 (0.978–1.014)
Systemic glucocorticoids	1.019 (0.991–1.048)

- The negative controls were null: low-TB-risk biologics 1.001 (0.985–1.018); levetiracetam 1.014 (0.985–1.043).
- Clustering by principal trust or using elective catchments barely changed the estimates.
- Lower-tier results were concordant.
- JAK inhibitors showed small inverse same-year estimates (0.993). These were not reproduced with previous-year or following-year exposure and are consistent with trends in uptake.

Table 3. Hospital medicines (SCMD, patient-year equivalents per 1,000 residents) and TB notifications, 2019–2024

Level	Drug group	Cross-sectional, adjusted (per SD)	Within-area, same year (per 10%)	Elasticity (same year)	Within-area, previous year (per 10%)	Previous year, SE clustered by principal trust	Following year (joint with previous)
UTLA	Active-TB treatment (pyrazinamide/ethambutol; positive control)	1.165 (1.026–1.324)	1.035 (1.013–1.058)	0.36	1.005 (0.995–1.014)	1.005 (0.995–1.014)	1.000 (0.988–1.012)
UTLA	Rifamycin/isoniazid (active and latent TB, other infections)	1.051 (1.032–1.284)	1.035 (1.021–1.050)	0.37	1.005 (0.993–1.018)	1.005 (0.993–1.018)	1.017 (0.994–1.040)
UTLA	TNF inhibitors	0.956 (0.905–1.010)	0.996 (0.979–1.013)	-0.04	0.989 (0.972–1.006)	0.989 (0.970–1.008)	0.978 (0.948–1.008)
UTLA	IL-6 inhibitors and abatacept	0.972 (0.923–1.024)	1.000 (0.990–1.010)	-0.00	1.000 (0.992–1.008)	1.000 (0.993–1.008)	1.001 (0.983–1.018)
UTLA	JAK inhibitors (rheumatology)	0.974 (0.916–1.035)	0.993 (0.987–0.999)	-0.07	0.994 (0.987–1.001)	0.994 (0.986–1.002)	0.998 (0.984–1.012)
UTLA	Rituximab	0.942 (0.885–1.003)	0.992 (0.979–1.006)	-0.08	0.998 (0.985–1.011)	0.998 (0.987–1.009)	1.007 (0.990–1.024)
UTLA	Calcineurin/mTOR inhibitors	1.047 (0.992–1.105)	0.994 (0.985–1.004)	-0.06	1.008 (0.996–1.020)	1.008 (0.996–1.020)	0.985 (0.966–1.005)
UTLA	Antiproliferatives	1.023 (0.968–1.081)	0.993 (0.978–1.008)	-0.07	0.996 (0.978–1.014)	0.996 (0.979–1.014)	0.993 (0.974–1.013)

Level	Drug group	Cross-sectional, adjusted (per SD)	Within-area, same year (per 10%)	Elasticity (same year)	Within-area, previous year (per 10%)	Previous year, SE clustered by principal trust	Following year (joint with previous)
UTLA	Systemic glucocorticoids	0.950 (0.912–0.990)	1.005 (0.977–1.033)	0.05	1.019 (0.991–1.048)	1.019 (0.987–1.053)	0.985 (0.935–1.037)
UTLA	Low-TB-risk biologics (negative control)	1.015 (0.962–1.072)	1.004 (0.990–1.019)	0.05	1.001 (0.985–1.018)	1.001 (0.983–1.020)	1.010 (0.981–1.039)
UTLA	Levetiracetam (negative control)	1.016 (0.968–1.066)	0.997 (0.975–1.020)	-0.03	1.014 (0.985–1.043)	1.014 (0.986–1.043)	0.985 (0.946–1.026)
LTLA	Active-TB treatment (pyrazinamide/ethambutol; positive control)	1.090 (1.011–1.175)	1.039 (1.017–1.062)	0.40	1.002 (0.993–1.012)	1.002 (0.993–1.012)	0.999 (0.987–1.011)
LTLA	Rifamycin/isoniazid (active and latent TB, other infections)	1.065 (1.000–1.135)	1.038 (1.025–1.051)	0.39	1.004 (0.992–1.016)	1.004 (0.992–1.016)	1.015 (0.995–1.036)
LTLA	TNF inhibitors	0.967 (0.925–1.010)	0.995 (0.979–1.010)	-0.06	0.992 (0.977–1.007)	0.992 (0.977–1.007)	0.975 (0.945–1.007)
LTLA	IL-6 inhibitors and abatacept	0.944 (0.905–0.983)	1.001 (0.991–1.011)	0.01	1.001 (0.993–1.009)	1.001 (0.993–1.009)	0.996 (0.979–1.012)
LTLA	JAK inhibitors (rheumatology)	0.948 (0.905–0.992)	0.994 (0.989–1.000)	-0.06	0.996 (0.989–1.002)	0.996 (0.989–1.002)	0.999 (0.985–1.013)
LTLA	Rituximab	0.971 (0.925–1.019)	0.991 (0.977–1.005)	-0.10	1.001 (0.986–1.016)	1.001 (0.988–1.014)	1.006 (0.988–1.023)
LTLA	Calcineurin/mTOR inhibitors	0.998 (0.955–1.043)	0.994 (0.985–1.003)	-0.07	1.009 (0.997–1.021)	1.009 (0.999–1.019)	0.983 (0.960–1.005)
LTLA	Antiproliferative	0.982 (0.935–1.030)	0.997 (0.983–1.012)	-0.03	0.997 (0.982–1.013)	0.997 (0.980–1.015)	0.993 (0.974–1.013)
LTLA	Systemic glucocorticoids	0.967 (0.934–1.002)	1.006 (0.980–1.034)	0.07	1.020 (0.993–1.047)	1.020 (0.990–1.051)	0.973 (0.927–1.022)

Level	Drug group	Cross-sectional, adjusted (per SD)	Within-area, same year (per 10%)	Elasticity (same year)	Within-area, previous year (per 10%)	Previous year, SE clustered by principal trust	Following year (joint with previous)
LTLA	Low-TB-risk biologics (negative control)	0.991 (0.948–1.036)	1.005 (0.991–1.019)	0.05	1.003 (0.988–1.018)	1.003 (0.987–1.019)	1.010 (0.983–1.038)
LTLA	Levetiracetam (negative control)	1.013 (0.975–1.052)	1.003 (0.981–1.025)	0.03	1.012 (0.988–1.037)	1.012 (0.989–1.035)	0.989 (0.951–1.027)

Minimum detectable effects and expected population effects

Primary care, oral glucocorticoids (lower-tier panel) - Minimum detectable effect (MDE): 6.3% per 10% increase in prescribing. - Expected effect under published relative risks and prevalence: 0.34% (Table 2). - Stratified by age and place of birth: 0.29%. - Based on UKHSA-recorded steroid-associated notifications: 0.055%. - The design could therefore detect only effects about 18–110 times larger than expected. - The MDE corresponds to a drug responsible for more than half of all notifications. - For the other candidate groups, MDEs exceeded expected effects by one to two orders of magnitude.

Hospital medicines - MDEs were smaller (0.9–4.7% per 10% increase), but so were expected effects. - TNF inhibitors (SCMD-derived prevalence 0.34%): - RR 4 without latent TB screening implies an expected change of 0.10% (MDE 25× larger). - A screening-attenuated RR implies 0.02% (MDE 113× larger). - Further attenuation by the positive-control elasticity widens these gaps (Table S3). - UKHSA-recorded biologic-associated notifications imply 0.098%.

Simulation (permutation-based, real notification counts; Table S4) - **Calibration.** - False-positive rate: 7.0% at lower-tier level (6.8% upper-tier). - Empirical null SD: 0.0158, against a real-data SE of 0.0218. The clustered SEs are therefore conservative for the estimate, but the slightly raised false-positive rate shows that permuted exposure histories still produce spurious associations with the real notification trends. - **Published glucocorticoid effect (RR 4.9).** Power 7.7% (upper-tier 9.0%). - **Larger individual relative risks.** Power 20.7% for RR 25 and 78.3% for RR 100. - **Direct population effects.** Power 21.0% for IRR 1.02 per 10%, 85.0% for 1.05 and 100.0% for 1.10. - **Effect acting through same-year prescribing** (analysed with previous-year exposure). Power 60.7% for 1.05 and 99.7% for 1.10.

Table 2. Minimum detectable effects (LTLA panel, residence apportionment, exposure t–1) versus expected population effects of a 10% increase in use

Drug group	Individual RR/OR	Prevalence of use	PAF	Expected change	MDE	MDE ÷ expected	Power	Minimum detectable PAF	RR for 80% power
Systemic oral glucocorticoids	4.9	0.9%	3.4%	0.34%	6.3%	18	3.6%	63%	191

Drug group	Individual RR/OR	Prevalence of use	PAF	Expected change	MDE	MDE ÷ expected	Power	Minimum detectable PAF	RR for 80% power
Inhaled corticosteroids	1.27	5.2%	1.4%	0.14%	7.0%	49	2.9%	70%	47
Conventional DMARDs	1.2	0.5%	0.1%	0.01%	2.8%	274	2.6%	28%	78
Proton pump inhibitors	1.28	14.2%	3.8%	0.38%	6.3%	16	3.7%	63%	13
Statins	0.6	12.8%	-5.4%	-0.54%	6.0%	11	4.5%	60%	not attainable
Metformin	0.51	4.4%	-2.2%	-0.22%	4.7%	21	3.4%	47%	not attainable

Expected changes use published individual-level relative risks (ORs for oral glucocorticoids and PPIs) and UK prevalence of use, assuming prevalence scales with prescribing and a homogeneous baseline risk (which favours detection). Power is one-sided power to detect the expected effect in the correct direction. Benchmarks: recorded cases (steroids), UKHSA 2024: 0.055%; recorded cases (biological therapy), UKHSA 2024: 0.098%; stratified by age and birthplace (oral corticosteroids, RR 4.9): 0.293%; homogeneous (oral corticosteroids, RR 4.9, prevalence 0.9%): 0.339%.

Regional analyses by place of birth and age

Non-UK-born notifications - Oral glucocorticoid prescribing in earlier years was not associated with notifications (lag 1-3: 0.92-1.02). - The falsification test failed: prescribing three years later predicted earlier notifications (1.49, 1.17-1.90; randomisation $p = 0.01$).

UK-born notifications - Prednisolone-equivalent mg: - Lagged estimates were imprecise positive values (lag 2: 1.30, 0.97-1.75; randomisation $p = 0.08$). - With lag and lead estimated jointly, lag 1.41 and lead 0.85 (difference $p = 0.04$; randomisation p for lag = 0.04) (Table S6). - Oral glucocorticoid items: null (lag 2 joint 1.22).

UK-born notifications at age ≥ 65 - Glucocorticoid prescribing was not associated with notifications. - The falsification test failed for glucocorticoid items (lead 2: 1.74, 0.99-3.07; randomisation $p = 0.03$). - The negative control failed in both directions: levothyroxine lag 1 1.23 (1.00-1.50; $p = 0.05$) and lead 1 1.37 (1.04-1.79; $p = 0.01$).

Interpretation of the one nominal association - The UK-born association with prednisolone mg is implausibly large. Under the model used for expected effects, a 10% increase in use can raise incidence by at most 10%. - Falsification tests and the negative control failed at similar magnitudes in the same regional design. - We therefore interpret it as residual confounding by regional trends in a nine-cluster design, not a drug effect.

Oral glucocorticoid prescribing and UK-born TB notifications by region, indexed to 2014 = 100

Prescribing: standard GP practices, per 1,000 residents. UK-born TB: UKHSA TB in England 2025, Supplementary Table 12 (LFS denominators; small annual counts in some regions).

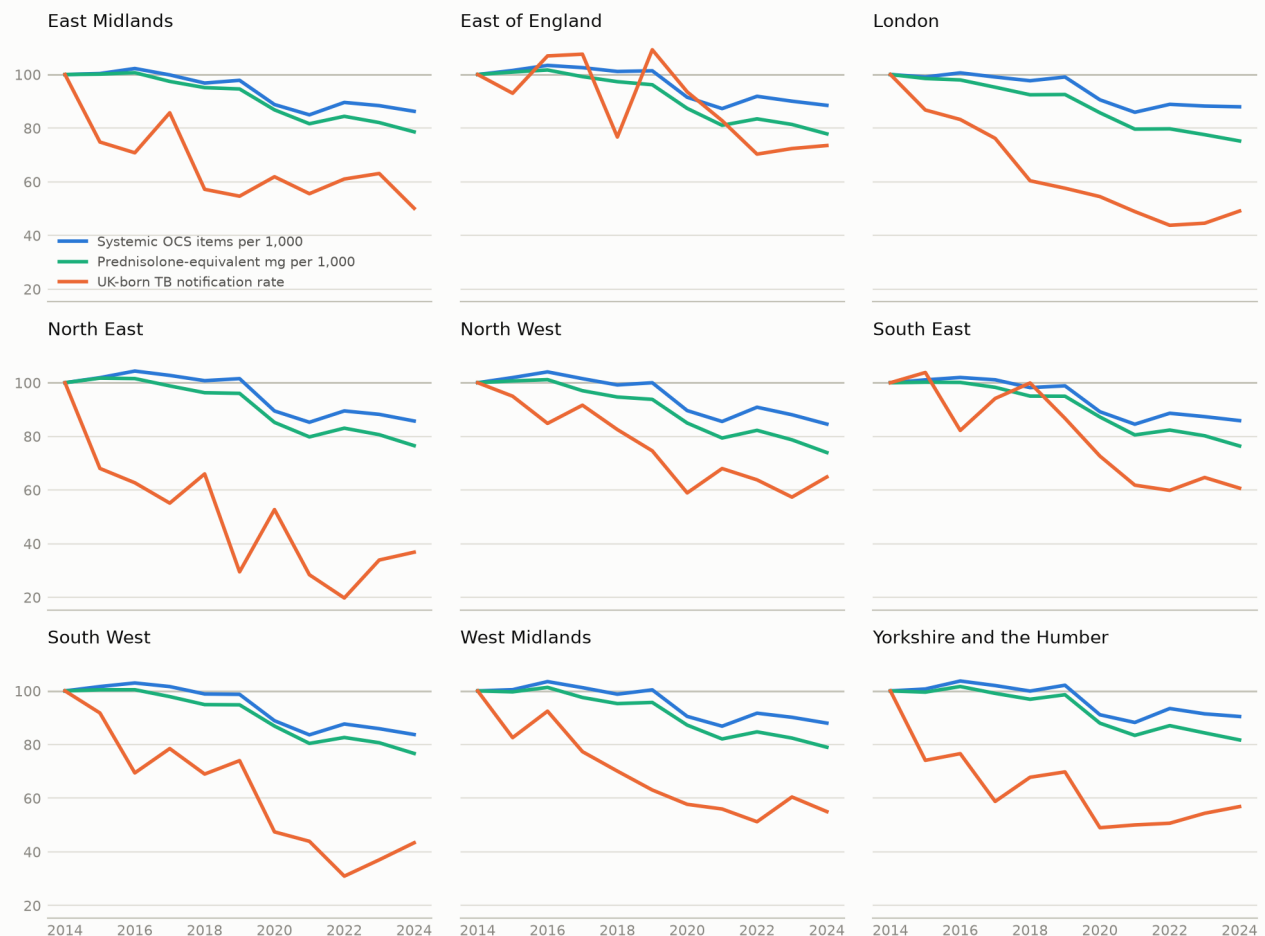


Figure 6: **Figure 6.** Oral glucocorticoid prescribing and UK-born TB notification rates by region, indexed to 2014 = 100.

Discussion

Principal findings

We linked open English data on primary care and hospital prescribing with TB notifications, using several ecological designs at two geographic scales. The linkage worked only partly.

- **The positive control.** Hospital active-TB treatment tracked notifications between and within areas, but even this disease-specific signal was recovered with roughly 60% attenuation.
- **Candidate drugs.** No medicine plausibly affecting TB risk was associated with subsequent notifications. This held for systemic oral glucocorticoids, inhaled corticosteroids, DMARDs, transplant immunosuppressants, TNF, IL-6 and JAK inhibitors, and rituximab.
- **Signs of confounding.**
 - The negative-control exposure showed inverse estimates of similar size to some candidate drugs.
 - Several falsification tests failed.
 - The one nominal regional association was implausibly large.

These patterns point to residual confounding by area-specific trends, not drug effects.

- **Power.** The analytic calculations and the simulation agreed: the designs could detect only effects one to two orders of magnitude larger than those implied by published relative risks or by UKHSA records of drug-associated TB.

Why population effects of these medicines are undetectable

Expected effects are small. The population effect of a medicine is roughly its attributable fraction scaled by the relative change in use.

- **Oral glucocorticoids.** Current use carries an odds ratio of about 5 [Jick 2006], but only about 1% of people use them at any time [van Staa 2000; Fardet 2011]. A 10% change in use should therefore change TB notifications by about 0.3%.
 - Stratifying by age and place of birth lowers this to 0.29%, because use is concentrated in older UK-born people with low baseline risk.
 - UKHSA recorded steroid-associated immunosuppression in 30 of 5,490 people notified in 2024 [UKHSA 2025]. Even allowing for under-recording, that implies 0.05–0.1%.
- **Biologics.** Latent TB screening before anti-TNF therapy has been standard in the UK since 2005 [BTS 2005] and reduced TB risk by about 78% in registry data [Carmona 2005; Gómez-Reino 2007]. Growth in biologic use during 2019–2024 therefore took place under screening, so the relative risk relevant to a marginal increase in use is small. Oral glucocorticoids rarely trigger screening.

The data carry little information. - **Little within-area variation.** Within-area variation in prescribing was very small (SD of log rate 0.04 for glucocorticoids, against 0.29 between areas), so a 10% change sits at the edge of the observed data. Fixed effects remove the between-area variation where most information lies [Gunasekara 2014]. - **Few areas.** Power in aggregate studies depends mainly on the number of areas [Sheppard 1996], which cannot be increased. - **Linkage attenuation.** The positive control shows that attenuation from apportionment and dilution would shrink any true effect further. - **Simulation.** Simulations that preserved the real error structure found power close to the false-positive rate for the published glucocorticoid effect.

Confounding, falsification and negative controls

Crude versus within-area associations. Crude cross-sectional associations were strongly inverse and were explained by country of birth and age. Within areas, residual confounding remained visible: - **Negative control.** Levothyroxine, which has no plausible effect on TB, showed inverse estimates (0.96–0.97) similar to metformin. - **Opposite-signed lag and lead.** Inhaled

corticosteroids and insulins had opposite-signed estimates when prescribing before and after the outcome year was estimated jointly. - **Likely cause.** Both patterns are expected when prescribing and notifications share area-specific trends, for example migration-driven changes in the age and origin of the population, changes in registration or diagnostic activity, and prescribing policy. Year fixed effects cannot remove these trends.

Ecological bias. Area-level adjustment cannot remove bias arising from effect modification or from differences in baseline risk between people within areas [Greenland 1989; Morgenstern 1995]. For example, steroid users are mostly older and UK-born, whereas most notifications are in younger people born abroad.

The UK-born association. Regionally, prednisolone dose was associated with UK-born TB when prescribing before and after was estimated jointly (randomisation $p = 0.04$). We do not interpret this as a drug effect: - An IRR of 1.41 per 10% increase exceeds the maximum possible under any relative risk (1.10) in the model used for expected effects. - The same nine-region design produced failed falsification tests for non-UK-born TB. - It also produced a failed negative control (levothyroxine) for older UK-born people. - UK-born TB fell by 43% between 2014 and 2022, reflecting social risk factors, transmission and demographic change [Davidson 2018; UKHSA 2025]. Coincident regional trends in steroid dosing are the most plausible explanation.

Protopathic bias. Glucocorticoids and antibacterials prescribed for undiagnosed TB, and glucocorticoids used in treating TB meningitis and pericarditis [Thwaites 2004], create reverse associations in the same year. This is why the primary exposure was prescribing in the previous year. The same bias probably inflates the individual-level estimates used to derive expected effects [Jick 2006]; if so, the true gap between expected and detectable effects is even wider.

Strengths and limitations

Strengths. - **Open and reproducible.** All data are public, and code and processed data are openly available. - **Exposure measurement.** - Primary care exposure was defined at presentation level, with dose measures, and apportioned by where registered patients live. - Hospital medicines were classified by mechanism using defined daily doses, with trust mergers resolved. - **Checks on validity.** - A disease-specific positive control, negative-control exposures and falsification tests estimated jointly. - Randomisation inference for the nine-region analyses. - Year-specific spatial diagnostics. - Power assessed both analytically and by simulation on the real data. - **Independent peer review.** Three independent reviews prompted substantial revision.

Limitations: exposure data. - **Prescribing measures.** Exposure was measured as prescribing volume per resident, not as people treated. - **Apportionment.** Apportioning by patient residence used annual April snapshots, with 2014 shares applied to 2011–2013. - **Pre-2014 data.** The 2011–2013 series come from a different release, and overlap-period concordance could not be assessed. National year-on-year changes were continuous across the 2013/2014 splice for most groups, but not for oral glucocorticoids (+3.3% at the splice against +0.6% and +1.0% either side) or all antibacterials (−0.1% against −4.5% and −5.6%). Restricting to EPD-era exposure did not change the conclusions (Table S1). - **Hospital catchments.** Hospital quantities were apportioned using admission-based catchments, which may misallocate specialist services. - **Hospital data capture.** Capture of homecare-delivered biologics in SCMD could not be verified. - **Hospital data period.** The hospital series spans only 2019–2024, including the pandemic.

Limitations: outcome and confounder data. - **Notifications, not incidence.** The outcome is notified TB, subject to diagnostic delay, residence assignment, the 2021 transition from ETS to NTBS, and 2020 disruption. - **Birthplace data.** TB by place of birth is published only by region, and UK-born population denominators by age are not published regionally. - **Unmeasured confounders.** - Social risk factors and diagnostic intensity. - The area-level share of recent arrivals. - Latent TB programme activity after 2019/20.

Limitations: design and analysis. - **Pre-specification.** Many analyses were added after

initial null results and are exploratory. - **Expected-effect inputs.** Relative risks for PPIs, statins and metformin come mainly from high-incidence settings, and prevalence inputs for hospital drugs are derived or illustrative. Neither would change the conclusions unless the true effects were an order of magnitude larger.

Implications

For research. - Open prescribing data are valuable for describing prescribing [Bacon 2020; OpenPrescribing 2026] but cannot estimate the effects of medicines on rare outcomes such as TB. - Ecological studies of rare outcomes should report the following alongside any associations: - minimum detectable effects; - a positive control with its elasticity; - negative controls; - falsification tests. - Causal questions need individual-level linked data analysed with target trial emulation (Box). Even national cohorts give wide intervals for modest relative risks [Peeling 2015]. The only drug-TB target trial emulation we identified concerned DPP-4 inhibitors [Chen 2025].

For TB programmes. - **The null results do not mean these drugs are safe.** Glucocorticoids, biologics and JAK inhibitors remain important for individual patients. Drug-associated TB is serious, often extrapulmonary, and largely preventable. - **Surveillance.** Monitoring is better served by more complete and detailed immunosuppression fields in NTBS than by ecological analysis of prescribing. Useful fields would record drug class, time since starting, and whether latent TB screening was done and treated. - **Audit.** The practical lever is auditing latent TB screening before biologics, JAK inhibitors and high-dose, long-term glucocorticoids. - **Aggregate data from UKHSA.** Aggregate UKHSA tables would allow population monitoring in the groups where drug-associated reactivation matters, namely: - notifications with recorded biologic or steroid immunosuppression; - notifications by area, year, place of birth, age and time since entry.

Conclusion

Open English prescribing and TB notification data can be linked, but with heavy attenuation. Ecological analyses of these data cannot detect the population-level effects of medicines on TB: plausible effects lie one to two orders of magnitude below what the designs can resolve, and the apparent associations that do arise reflect confounding by migration and area-specific trends.

Box. Individual-level target trial emulation (example: oral glucocorticoids and TB)

Data - CPRD Aurum (primary care prescribing with dose and quantity). - Hospital Episode Statistics (admitted patient care and outpatients) and ONS deaths. - Small-area deprivation. - A bespoke linkage to the UKHSA National TB Surveillance System, which provides notification-validated TB, site of disease, culture confirmation, and country of birth and year of entry. - Hospital-only drugs (biologics, JAK inhibitors) are not captured in CPRD. For these, use NHS England high-cost drugs data (e.g. via OpenSAFELY) or disease registries (BSRBR-RA, BADBIR, UK IBD Registry) with the same TB linkage.

Eligibility - Adults aged ≥ 18 years with ≥ 12 months' registration. - No previous TB, TB or latent TB treatment, or oral glucocorticoid use in the previous 12 months. - No HIV or solid organ transplant. - Defined indication cohorts (e.g. polymyalgia rheumatica/giant cell arteritis, rheumatoid arthritis, COPD, asthma, inflammatory bowel disease).

Treatment strategies - Initiate and sustain oral glucocorticoids for ≥ 3 months, by prednisolone-equivalent dose (< 7.5 , 7.5 - < 15 , ≥ 15 mg/day), versus no initiation. - An active comparator where available.

Assignment and time zero - Sequential monthly nested trials. - Clone-censor-weight methods for sustained-use and dose strategies.

Outcome - Incident active TB: the first of an NTBS notification, a HES diagnosis (ICD-10 A15-A19), or a CPRD diagnosis with treatment. - Secondary: pulmonary vs extrapulmonary, culture-confirmed TB, TB death.

Follow-up - Up to 5 years, or until death, deregistration or end of linkage.

Estimands and analysis - Intention-to-treat and per-protocol cumulative incidence differences and ratios at 1, 2 and 5 years. - Pooled logistic regression with inverse probability of treatment and censoring weights.

Confounders - Age, sex, ethnicity, country of birth and time since entry, deprivation. - Diabetes, chronic kidney disease, smoking, alcohol, BMI. - TB contact or latent TB screening codes. - Indication severity, healthcare use, co-prescribed immunosuppressants.

Effect modification and bias checks - Effect modification by country of birth, specified in advance. - Negative-control exposure (levothyroxine initiation) and negative-control outcome. - E-values.

Feasibility - Baseline TB incidence in older UK-born adults is about 2-5 per 100,000 person-years. Tens of thousands of long-term initiators followed for several years would yield only tens of TB events, so the individual-level study is itself constrained by sample size.

Data and code availability

All inputs are publicly available (NHSBSA Open Data Portal: English Prescribing Dataset and Secondary Care Medicines Data; NHS Digital practice-level prescribing and registered patients by LSOA; UKHSA TB reports and Fingertips; OHID acute trust catchment populations; ONS/Nomis; Home Office asylum statistics; NHS England ODS; MHCLG English Indices of Deprivation). Code, processed datasets and outputs: <https://github.com/drcjar/tb-prescribing-england>.

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Each journal entry was checked against PubMed metadata (via DOI→PMID conversion or citation lookup) and Crossref DOI metadata. Each DOI below resolves to the stated article. The grey literature was checked on GOV.UK, in the report PDF, or through search results where a site blocks automated access. Corrections relative to references_v3.md are listed at the end.

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Corrections made relative to references_v3.md

- Thomas HL 2018: the title was wrong. The actual title is “Reduction in tuberculosis incidence in the UK from 2011 to 2015: a population-based study”. Volume and pages (73:769–75) were correct.
- Költringer 2023: the placeholder title was replaced, and the article number is 337, not 311.
- Castellana 2019: the full title ends “...meta-analysis of non-randomized studies”.
- Song 2019: the title was wrong. The actual title is “The association between proton pump inhibitor use and the risk of tuberculosis: a case-control study”.
- Hudson 2025: there are only two authors (Hudson SM, Hudson C), so “et al.” was removed. The print issue is 2026;33(1):1–8, epub 2025.
- Placeholder titles were replaced for Chen (BMC Med 2025;23:587), Hermans, Richards (Sci Rep 2025;15:18089), Sheppard (Stat Med 1996;15:1849–58, authors Sheppard, Prentice, Rossing) and Venkatesan (2025;11:e64788).
- Tuberculosis in England 2021 report: it was published by the UK Health Security Agency, not Public Health England. The document and its suggested citation both name UKHSA, and UKHSA replaced PHE on 1 October 2021, before the report was added on 28 October 2021.
- Added the Santos Silva DOI, all PMIDs and issue numbers, and the NICE NG33 publication and update dates.

Could not be fully verified

- NICE NG33: nice.org.uk blocks automated access (HTTP 403). The 16 February 2024 last-updated date comes from search results only, and the LTBI recommendation wording must be checked by hand.
- The OpenPrescribing FAQ page could not be fetched directly. Its content was confirmed through search results only.
- UKHSA 2025 report: which supplementary tables are numbered 5, 12 and 22, and the “TB regional reports 2024” supplementary data, were not individually checked.
- The full author lists of the “et al.” entries (Chen, Költringer, Morrison, Richards, Venkatesan) were not expanded. The PubMed records list 9, 5, 10, 5 and 14 authors respectively.

Supplementary tables

Table S1. Sensitivity analyses, LTLA panel (residence apportionment): IRR per 10% increase (95% CI)

Drug group	primary (t-1)	FE only (t-1)	prior 3 years (t-1)	+ area trends (t-1)	population co-vari-ate (t-1)	+ LTBI pro-gramme (t-1)	+ asy-lum sup-port (t-1)	exclude COVID years (t-1)	outcome years 2018-2024 (t-1)	EPD expo-sure only (t-1)	ADQ mea-sure (t-1)
Antituberculosis (descriptive)	1.000 (0.997-1.003)	1.002 (0.998-1.005)	1.001 (0.997-1.005)	0.999 (0.996-1.003)	1.000 (0.997-1.004)	1.000 (0.997-1.004)	1.001 (0.998-1.004)	1.000 (0.997-1.004)	1.001 (0.998-1.004)	1.001 (0.998-1.004)	—
Systemic oral glucocorticoids	0.982 (0.941-1.025)	0.972 (0.925-1.021)	0.988 (0.941-1.037)	0.999 (0.945-1.057)	0.984 (0.943-1.028)	0.983 (0.941-1.028)	0.993 (0.949-1.039)	0.990 (0.947-1.035)	0.986 (0.944-1.030)	0.991 (0.950-1.033)	—
Oral hydrocortisone	1.002 (0.990-1.014)	0.996 (0.982-1.012)	1.000 (0.984-1.016)	1.002 (0.984-1.019)	1.003 (0.991-1.016)	1.002 (0.990-1.014)	0.996 (0.984-1.009)	1.002 (0.987-1.017)	0.994 (0.978-1.011)	0.998 (0.987-1.010)	—
Oral dexamethasone	1.004 (0.994-1.015)	1.006 (0.994-1.018)	1.009 (0.994-1.025)	0.996 (0.986-1.007)	1.004 (0.994-1.015)	1.005 (0.994-1.015)	1.003 (0.993-1.014)	1.009 (0.998-1.020)	0.998 (0.986-1.010)	1.004 (0.994-1.014)	—
Inhaled corticosteroids	0.981 (0.935-1.029)	0.952 (0.905-1.001)	0.999 (0.947-1.053)	1.053 (1.003-1.105)	0.984 (0.935-1.034)	0.982 (0.936-1.030)	0.964 (0.921-1.008)	0.977 (0.930-1.026)	0.948 (0.902-0.996)	0.970 (0.929-1.013)	0.975 (0.952-0.999)
Conventional DMARDs	1.003 (0.984-1.023)	1.005 (0.985-1.026)	1.004 (0.984-1.026)	1.022 (0.985-1.060)	1.003 (0.984-1.022)	1.003 (0.984-1.022)	0.998 (0.980-1.016)	1.006 (0.987-1.026)	1.003 (0.979-1.026)	1.001 (0.985-1.018)	—
Transplant immunosuppressants	0.999 (0.993-1.005)	0.997 (0.991-1.003)	1.000 (0.993-1.007)	1.001 (0.993-1.008)	0.999 (0.993-1.005)	0.999 (0.993-1.005)	0.999 (0.993-1.005)	1.001 (0.995-1.008)	1.001 (0.994-1.008)	1.000 (0.995-1.006)	—
Proton pump inhibitors	1.009 (0.967-1.054)	0.975 (0.930-1.022)	1.006 (0.960-1.055)	0.988 (0.932-1.048)	1.011 (0.965-1.059)	1.014 (0.970-1.060)	0.997 (0.952-1.044)	1.006 (0.961-1.054)	0.947 (0.888-1.010)	1.001 (0.958-1.046)	1.000 (0.938-1.065)
Statins	1.000 (0.960-1.041)	0.974 (0.933-1.017)	0.995 (0.952-1.041)	0.996 (0.944-1.051)	0.999 (0.959-1.041)	1.005 (0.964-1.047)	1.009 (0.965-1.054)	0.991 (0.951-1.034)	0.997 (0.936-1.063)	1.009 (0.969-1.051)	1.002 (0.963-1.043)
Metformin	0.969 (0.938-1.001)	0.953 (0.921-0.986)	0.968 (0.933-1.004)	0.966 (0.925-1.009)	0.970 (0.938-1.003)	0.974 (0.942-1.008)	0.977 (0.942-1.014)	0.965 (0.935-0.997)	0.985 (0.931-1.042)	0.977 (0.944-1.011)	0.991 (0.941-1.044)
Insulins	0.981 (0.945-1.019)	0.980 (0.941-1.020)	0.989 (0.948-1.030)	1.042 (0.980-1.109)	0.986 (0.944-1.029)	0.983 (0.947-1.022)	0.980 (0.936-1.027)	0.979 (0.940-1.020)	0.982 (0.915-1.054)	0.985 (0.944-1.028)	—
Fluoroquinolones	1.010 (0.997-1.023)	1.010 (0.995-1.024)	1.011 (0.998-1.025)	1.008 (0.994-1.023)	1.011 (0.998-1.024)	1.009 (0.997-1.022)	1.008 (0.995-1.021)	1.006 (0.992-1.020)	0.995 (0.976-1.015)	1.006 (0.994-1.019)	1.004 (0.992-1.016)

Drug group	primary (t-1)	FE only (t-1)	prior 3 years (t-1)	+ area trends (t-1)	population + co-vari-ate (t-1)	+ LTBI pro-gramme (t-1)	+ asy-lum sup-port (t-1)	exclude COVID years (t-1)	outcome years 2018-2024 (t-1)	EPD expo-sure only (t-1)	ADQ mea-sure (t-1)
All an-tibac-teri-als	1.012 (0.971-1.054)	1.009 (0.961-1.058)	1.023 (0.980-1.069)	0.995 (0.955-1.037)	1.015 (0.976-1.056)	1.007 (0.965-1.051)	1.005 (0.964-1.046)	1.030 (0.984-1.078)	0.957 (0.914-1.002)	1.007 (0.968-1.048)	1.007 (0.970-1.045)
Vitamin D	1.008 (0.989-1.029)	1.011 (0.988-1.034)	1.009 (0.988-1.030)	0.986 (0.962-1.010)	1.010 (0.991-1.030)	1.006 (0.985-1.027)	1.007 (0.984-1.030)	1.019 (0.998-1.040)	0.989 (0.958-1.020)	1.006 (0.986-1.027)	—
Levothyroxine (negative control)	0.970 (0.938-1.003)	0.948 (0.919-0.978)	0.968 (0.936-1.001)	0.980 (0.930-1.033)	0.968 (0.936-1.002)	0.975 (0.940-1.011)	0.977 (0.941-1.014)	0.967 (0.930-1.006)	0.980 (0.929-1.032)	0.978 (0.943-1.014)	0.964 (0.916-1.014)

Table S2. Distributed lag model (t, t-1, t-2 in one model), LTLA residence

Drug group	t (same year)	t-1	t-2
Antituberculosis (descriptive)	0.999 (0.996-1.002)	1.001 (0.998-1.004)	1.000 (0.997-1.003)
Systemic oral glucocorticoids	0.966 (0.924-1.009)	1.003 (0.952-1.057)	1.007 (0.963-1.053)
Oral hydrocortisone	1.009 (0.991-1.027)	0.993 (0.973-1.014)	1.002 (0.983-1.022)
Oral dexamethasone	1.001 (0.991-1.011)	1.004 (0.994-1.013)	1.001 (0.991-1.011)
Inhaled corticosteroids	0.896 (0.849-0.946)	1.042 (0.966-1.123)	1.034 (0.969-1.104)
Conventional DMARDs	1.005 (0.968-1.043)	0.994 (0.936-1.056)	1.006 (0.964-1.049)
Transplant immunosuppressants	1.000 (0.990-1.010)	0.996 (0.983-1.009)	1.004 (0.995-1.013)
Proton pump inhibitors	0.994 (0.924-1.069)	1.048 (0.969-1.134)	0.965 (0.893-1.044)
Statins	1.013 (0.942-1.089)	1.022 (0.942-1.110)	0.965 (0.898-1.038)
Metformin	1.005 (0.943-1.071)	0.995 (0.925-1.071)	0.968 (0.915-1.024)
Insulins	0.931 (0.872-0.995)	1.033 (0.954-1.118)	1.006 (0.951-1.064)
Fluoroquinolones	1.002 (0.986-1.019)	1.002 (0.985-1.020)	1.007 (0.992-1.022)
All antibacterials	1.020 (0.985-1.056)	0.972 (0.928-1.017)	1.030 (0.987-1.075)
Vitamin D	1.011 (0.976-1.048)	0.991 (0.947-1.038)	1.008 (0.975-1.041)
Levothyroxine (negative control)	0.987 (0.923-1.056)	1.025 (0.946-1.110)	0.955 (0.889-1.026)

Table S3. Hospital medicines: minimum detectable effects (previous-year exposure) versus illustrative expected effects

Level	Drug group	Scenario (illustrative RR)	Prevalence (SCMD patient-years)	Expected change per 10%	MDE	MDE ÷ expected	MDE ÷ expected, attenuated by positive-control elasticity
UTLA	TNF inhibitors	RR 4 (unscreened)	0.34%	0.100%	2.5%	25	70
UTLA	TNF inhibitors	RR 1.7 (with LTBI screening, ~78% reduction)	0.34%	0.022%	2.5%	113	313
UTLA	IL-6 inhibitors and abat-accept	RR 2	0.02%	0.002%	1.2%	507	1403
UTLA	JAK inhibitors (rheumatology)	RR 2	0.04%	0.004%	1.0%	229	633
UTLA	Rituximab	RR 1.5	0.09%	0.005%	1.9%	416	1152
UTLA	Calcineurin/mTOR inhibitors	RR 10 (solid organ transplant)	0.07%	0.067%	1.7%	26	71
UTLA	Antiproliferatives	RR 2	0.16%	0.016%	2.6%	161	447
UTLA	Systemic glucocorticoids	RR 4.9	0.46%	0.177%	4.1%	23	62
LTLA	TNF inhibitors	RR 4 (unscreened)	0.34%	0.100%	2.2%	22	54
LTLA	TNF inhibitors	RR 1.7 (with LTBI screening, ~78% reduction)	0.34%	0.022%	2.2%	98	244
LTLA	IL-6 inhibitors and abat-accept	RR 2	0.02%	0.002%	1.2%	517	1279
LTLA	JAK inhibitors (rheumatology)	RR 2	0.04%	0.004%	0.9%	211	521
LTLA	Rituximab	RR 1.5	0.09%	0.005%	2.2%	471	1166

Level	Drug group	Scenario (illustrative RR)	Prevalence (SCMD patient-years)	Expected change per 10%	MDE	MDE ÷ expected	MDE ÷ expected, attenuated by positive-control elasticity
LTLA	Calcineurin/mTOR inhibitors	RR 10 (solid organ transplant)	0.07%	0.067%	1.7%	25	61
LTLA	Antiproliferatives	RR 2	0.16%	0.016%	2.2%	138	343
LTLA	Systemic glucocorticoids	RR 4.9	0.46%	0.177%	3.8%	21	53

Table S4. Permutation-based power simulation (systemic oral glucocorticoids, exposure t–1)

Level	Scenario	Replicates	Two-sided rejection	Power, correct direction (95% MC interval)	Median estimated IRR per 10%	Null empirical SD	Real-data SE
LTLA	No effect	500	7.0%	3.2% (±1.5)	1.001	0.0158	0.0218
LTLA	Published RR 4.9	300	7.7%	4.3% (±2.3)	1.005	0.0158	0.0218
LTLA	RR 25	300	20.7%	20.3% (±4.6)	1.017	0.0158	0.0218
LTLA	RR 100	300	78.3%	78.3% (±4.7)	1.046	0.0158	0.0218
LTLA	IRR 1.02 per 10%	300	21.0%	21.0% (±4.6)	1.019	0.0158	0.0218
LTLA	IRR 1.05 per 10%	300	85.0%	85.0% (±4.0)	1.052	0.0158	0.0218
LTLA	IRR 1.10 per 10%	300	100.0%	100.0% (±0.0)	1.101	0.0158	0.0218
LTLA	IRR 1.05 per 10%, acting via concurrent year	300	60.7%	60.7% (±5.5)	1.037	0.0158	0.0218
LTLA	IRR 1.10 per 10%, acting via concurrent year	300	99.7%	99.7% (±0.7)	1.080	0.0158	0.0218
UTLA	No effect	500	6.8%	3.8% (±1.7)	1.000	0.0166	0.0250
UTLA	Published RR 4.9	300	9.0%	8.0% (±3.1)	1.003	0.0166	0.0250

Level	Scenario	Replicates	Two-sided rejection	Power, correct direction (95% MC interval)	Median estimated IRR per 10%	Null empirical SD	Real-data SE
UTLA	RR 25	300	20.3%	19.7% (± 4.5)	1.017	0.0166	0.0250
UTLA	RR 100	300	71.7%	71.7% (± 5.1)	1.046	0.0166	0.0250
UTLA	IRR 1.02 per 10%	300	24.7%	24.7% (± 4.9)	1.023	0.0166	0.0250
UTLA	IRR 1.05 per 10%	300	81.3%	81.3% (± 4.4)	1.050	0.0166	0.0250
UTLA	IRR 1.10 per 10%	300	100.0%	100.0% (± 0.0)	1.100	0.0166	0.0250
UTLA	IRR 1.05 per 10%, acting via concurrent year	300	64.0%	64.0% (± 5.4)	1.042	0.0166	0.0250
UTLA	IRR 1.10 per 10%, acting via concurrent year	300	99.3%	99.3% (± 0.9)	1.085	0.0166	0.0250

Table S5. Regional models (9 regions): TB notifications by place of birth and age vs primary care prescribing

Outcome	Exposure	Analysis	Lag (years; negative = lead)	IRR per 10% (t(8) 95% CI)	p (cluster t8)	p (randomisation)
TB, UK born	oral_glucocorticoids	lagged exposure	1	1.18 (0.87–1.60)	0.238	0.284
TB, UK born	oral_glucocorticoids	lagged exposure	2	1.18 (0.79–1.77)	0.376	0.354
TB, UK born	oral_glucocorticoids	lagged exposure	3	1.07 (0.64–1.76)	0.778	0.768
TB, UK born	oral_glucocorticoids	lagged, excluding 2020-21	1	1.20 (0.86–1.66)	0.243	0.226
TB, UK born	oral_glucocorticoids	lagged, excluding 2020-21	2	1.19 (0.79–1.80)	0.362	0.342
TB, UK born	oral_glucocorticoids	lagged, excluding 2020-21	3	1.08 (0.65–1.80)	0.743	0.762
TB, UK born	oral_glucocorticoids	lead (falsification)	-1	1.20 (0.76–1.90)	0.393	0.262
TB, UK born	oral_glucocorticoids	lead (falsification)	-2	1.07 (0.65–1.76)	0.769	0.706
TB, UK born	oral_glucocorticoids	lead (falsification)	-3	1.11 (0.82–1.49)	0.454	0.576

Outcome	Exposure	Analysis	Lag (years; negative = lead)	IRR per 10% (t(8) 95% CI)	p (cluster t8)	p (randomi- sation)
TB, UK born	mg_pred_equivalent	lagged exposure	1	1.23 (0.95–1.58)	0.102	0.132
TB, UK born	mg_pred_equivalent	lagged exposure	2	1.30 (0.97–1.75)	0.076	0.078
TB, UK born	mg_pred_equivalent	lagged exposure	3	1.33 (0.93–1.89)	0.103	0.134
TB, UK born	mg_pred_equivalent	lagged, excluding 2020-21	1	1.27 (0.93–1.72)	0.113	0.086
TB, UK born	mg_pred_equivalent	lagged, excluding 2020-21	2	1.30 (0.97–1.75)	0.073	0.072
TB, UK born	mg_pred_equivalent	lagged, excluding 2020-21	3	1.30 (0.89–1.89)	0.149	0.212
TB, UK born	mg_pred_equivalent	lead (falsifi- cation)	-1	1.15 (0.84–1.56)	0.337	0.326
TB, UK born	mg_pred_equivalent	lead (falsifi- cation)	-2	0.89 (0.71–1.13)	0.303	0.546
TB, UK born	mg_pred_equivalent	lead (falsifi- cation)	-3	0.93 (0.84–1.03)	0.121	0.662
TB, UK born	levothyroxine	lagged exposure	1	0.94 (0.84–1.05)	0.246	0.588
TB, UK born	levothyroxine	lagged exposure	2	0.89 (0.79–1.00)	0.050	0.390
TB, UK born	levothyroxine	lagged exposure	3	0.89 (0.76–1.03)	0.097	0.390
TB, UK born	levothyroxine	lagged, excluding 2020-21	1	0.94 (0.83–1.06)	0.270	0.572
TB, UK born	levothyroxine	lagged, excluding 2020-21	2	0.89 (0.78–1.02)	0.079	0.404
TB, UK born	levothyroxine	lagged, excluding 2020-21	3	0.89 (0.75–1.05)	0.136	0.404
TB, UK born	levothyroxine	lead (falsifi- cation)	-1	0.92 (0.80–1.07)	0.257	0.488
TB, UK born	levothyroxine	lead (falsifi- cation)	-2	0.95 (0.80–1.14)	0.560	0.654
TB, UK born	levothyroxine	lead (falsifi- cation)	-3	0.97 (0.80–1.16)	0.675	0.694
TB, Non-UK born	oral_glucocorticoids	lagged exposure	1	0.92 (0.70–1.20)	0.478	0.588
TB, Non-UK born	oral_glucocorticoids	lagged exposure	2	0.97 (0.80–1.18)	0.737	0.834
TB, Non-UK born	oral_glucocorticoids	lagged exposure	3	1.02 (0.81–1.30)	0.820	0.854

Outcome	Exposure	Analysis	Lag (years; negative = lead)	IRR per 10% (t(8) 95% CI)	p (cluster t8)	p (randomi- sation)
TB, Non-UK born	oral_glucocorticoids	lagged, excluding 2020-21	1	0.95 (0.70–1.28)	0.678	0.732
TB, Non-UK born	oral_glucocorticoids	lagged, excluding 2020-21	2	1.00 (0.82–1.22)	0.968	0.978
TB, Non-UK born	oral_glucocorticoids	lagged, excluding 2020-21	3	1.06 (0.83–1.35)	0.606	0.700
TB, Non-UK born	oral_glucocorticoids	lead (falsifi- cation)	-1	1.15 (0.91–1.46)	0.216	0.424
TB, Non-UK born	oral_glucocorticoids	lead (falsifi- cation)	-2	1.37 (1.01–1.86)	0.046	0.112
TB, Non-UK born	oral_glucocorticoids	lead (falsifi- cation)	-3	1.49 (1.17–1.90)	0.005	0.010
TB, Non-UK born	mg_pred_equivalents	lagged exposure	1	0.97 (0.76–1.23)	0.745	0.814
TB, Non-UK born	mg_pred_equivalents	lagged exposure	2	1.06 (0.92–1.22)	0.393	0.650
TB, Non-UK born	mg_pred_equivalents	lagged exposure	3	1.11 (0.94–1.31)	0.185	0.444
TB, Non-UK born	mg_pred_equivalents	lagged, excluding 2020-21	1	1.00 (0.78–1.29)	0.988	0.992
TB, Non-UK born	mg_pred_equivalents	lagged, excluding 2020-21	2	1.06 (0.92–1.22)	0.395	0.662
TB, Non-UK born	mg_pred_equivalents	lagged, excluding 2020-21	3	1.13 (0.94–1.36)	0.176	0.384
TB, Non-UK born	mg_pred_equivalents	lead (falsifi- cation)	-1	1.16 (0.90–1.51)	0.208	0.356
TB, Non-UK born	mg_pred_equivalents	lead (falsifi- cation)	-2	1.32 (0.94–1.86)	0.096	0.122
TB, Non-UK born	mg_pred_equivalents	lead (falsifi- cation)	-3	1.28 (1.03–1.60)	0.033	0.124
TB, Non-UK born	levothyroxine	lagged exposure	1	1.02 (0.89–1.15)	0.786	0.868
TB, Non-UK born	levothyroxine	lagged exposure	2	1.01 (0.88–1.17)	0.843	0.908

Outcome	Exposure	Analysis	Lag (years; negative = lead)	IRR per 10% (t(8) 95% CI)	p (cluster t8)	p (randomi- sation)
TB, Non-UK born	levothyroxine	lagged exposure	3	1.01 (0.88-1.16)	0.859	0.914
TB, Non-UK born	levothyroxine	lagged, excluding 2020-21	1	1.02 (0.89-1.16)	0.762	0.864
TB, Non-UK born	levothyroxine	lagged, excluding 2020-21	2	1.01 (0.88-1.17)	0.840	0.902
TB, Non-UK born	levothyroxine	lagged, excluding 2020-21	3	1.01 (0.88-1.16)	0.856	0.920
TB, Non-UK born	levothyroxine	lead (falsifi- cation)	-1	1.00 (0.85-1.16)	0.961	0.988
TB, Non-UK born	levothyroxine	lead (falsifi- cation)	-2	1.02 (0.88-1.18)	0.806	0.920
TB, Non-UK born	levothyroxine	lead (falsifi- cation)	-3	1.09 (0.91-1.29)	0.300	0.550
TB, UK born aged 65+	oral_glucocorticoids	lagged exposure	1	1.04 (0.71-1.51)	0.823	0.868
TB, UK born aged 65+	oral_glucocorticoids	lagged exposure	2	1.02 (0.68-1.54)	0.894	0.894
TB, UK born aged 65+	oral_glucocorticoids	lagged exposure	3	0.86 (0.44-1.68)	0.618	0.434
TB, UK born aged 65+	oral_glucocorticoids	lagged, excluding 2020-21	1	1.11 (0.77-1.59)	0.530	0.624
TB, UK born aged 65+	oral_glucocorticoids	lagged, excluding 2020-21	2	1.10 (0.73-1.66)	0.615	0.592
TB, UK born aged 65+	oral_glucocorticoids	lagged, excluding 2020-21	3	0.92 (0.50-1.68)	0.757	0.682
TB, UK born aged 65+	oral_glucocorticoids	lead (falsifi- cation)	-1	1.29 (0.82-2.02)	0.228	0.204
TB, UK born aged 65+	oral_glucocorticoids	lead (falsifi- cation)	-2	1.74 (0.99-3.07)	0.055	0.028
TB, UK born aged 65+	oral_glucocorticoids	lead (falsifi- cation)	-3	1.55 (0.96-2.49)	0.066	0.070
TB, UK born aged 65+	mg_pred_equivalent	lagged exposure	1	1.06 (0.75-1.49)	0.707	0.684

Outcome	Exposure	Analysis	Lag (years; negative = lead)	IRR per 10% (t(8) 95% CI)	p (cluster t8)	p (randomi- sation)
TB, UK born aged 65+	mg_pred_equivalent	lagged exposure	2	1.10 (0.79–1.53)	0.524	0.548
TB, UK born aged 65+	mg_pred_equivalent	lagged exposure	3	0.97 (0.54–1.73)	0.909	0.882
TB, UK born aged 65+	mg_pred_equivalent	lagged, excluding 2020-21	1	1.14 (0.77–1.69)	0.463	0.424
TB, UK born aged 65+	mg_pred_equivalent	lagged, excluding 2020-21	2	1.12 (0.76–1.64)	0.527	0.418
TB, UK born aged 65+	mg_pred_equivalent	lagged, excluding 2020-21	3	0.93 (0.49–1.79)	0.809	0.722
TB, UK born aged 65+	mg_pred_equivalent	lead (falsifi- cation)	-1	1.29 (0.86–1.91)	0.185	0.150
TB, UK born aged 65+	mg_pred_equivalent	lead (falsifi- cation)	-2	1.59 (0.88–2.87)	0.107	0.086
TB, UK born aged 65+	mg_pred_equivalent	lead (falsifi- cation)	-3	1.31 (0.81–2.14)	0.233	0.252
TB, UK born aged 65+	levothyroxine	lagged exposure	1	1.23 (1.00–1.50)	0.051	0.050
TB, UK born aged 65+	levothyroxine	lagged exposure	2	1.16 (0.95–1.42)	0.126	0.226
TB, UK born aged 65+	levothyroxine	lagged exposure	3	1.07 (0.90–1.26)	0.415	0.584
TB, UK born aged 65+	levothyroxine	lagged, excluding 2020-21	1	1.22 (1.00–1.50)	0.053	0.054
TB, UK born aged 65+	levothyroxine	lagged, excluding 2020-21	2	1.18 (0.97–1.44)	0.090	0.142
TB, UK born aged 65+	levothyroxine	lagged, excluding 2020-21	3	1.10 (0.92–1.32)	0.239	0.398
TB, UK born aged 65+	levothyroxine	lead (falsifi- cation)	-1	1.37 (1.04–1.79)	0.028	0.008
TB, UK born aged 65+	levothyroxine	lead (falsifi- cation)	-2	1.20 (0.84–1.72)	0.276	0.302
TB, UK born aged 65+	levothyroxine	lead (falsifi- cation)	-3	1.29 (0.83–2.00)	0.223	0.106

Table S6. Regional models: lag 2 and lead 2 estimated jointly

Outcome	Exposure	Years	IRR lag (per 10%)	IRR lead (per 10%)	Lag ÷ lead	p difference	p randomi- sation (lag)
TB, UK born	oral_glucocorticoids	2013- 2022	1.22	0.94	1.29	0.239	0.324
TB, UK born	mg_pred_equivalents	2013- 2022	1.41	0.85	1.66	0.037	0.040
TB, UK born	levothyroxine	2013- 2022	0.82	1.00	0.82	0.564	0.408
TB, Non-UK born	oral_glucocorticoids	2013- 2022	0.94	1.40	0.67	0.111	0.742
TB, Non-UK born	mg_pred_equivalents	2013- 2022	1.09	1.27	0.86	0.498	0.628
TB, Non-UK born	levothyroxine	2013- 2022	0.87	1.07	0.81	0.557	0.466
TB, UK born aged 65+	oral_glucocorticoids	2013- 2022	0.81	1.78	0.46	0.079	0.372
TB, UK born aged 65+	mg_pred_equivalents	2013- 2022	1.01	1.53	0.66	0.302	0.968
TB, UK born aged 65+	levothyroxine	2013- 2022	0.92	1.14	0.81	0.546	0.790

Table S7. Hospital medicines: share of DDD from trusts matched to catchments directly, via successor trusts, or unmatched

Drug group	Year	Direct	Successor	Unmatched
Antiproliferatives	2019	90.9%	8.4%	0.7%
Antiproliferatives	2020	93.8%	4.8%	1.4%
Antiproliferatives	2021	96.4%	2.4%	1.2%
Antiproliferatives	2022	97.7%	0.7%	1.6%
Antiproliferatives	2023	97.6%	0.5%	1.9%
Antiproliferatives	2024	97.7%	0.4%	1.9%
Active-TB treatment (pyrazi- namide/ethambutol; positive control)	2019	91.3%	8.3%	0.4%
Active-TB treatment (pyrazi- namide/ethambutol; positive control)	2020	94.6%	5.1%	0.3%

Drug group	Year	Direct	Successor	Unmatched
Active-TB treatment (pyrazinamide/ethambutol; positive control)	2021	97.4%	2.5%	0.1%
Active-TB treatment (pyrazinamide/ethambutol; positive control)	2022	98.5%	1.4%	0.1%
Active-TB treatment (pyrazinamide/ethambutol; positive control)	2023	98.9%	1.0%	0.1%
Active-TB treatment (pyrazinamide/ethambutol; positive control)	2024	98.3%	0.9%	0.7%
Rifamycin/isoniazid (active and latent TB, other infections)	2019	90.6%	9.0%	0.3%
Rifamycin/isoniazid (active and latent TB, other infections)	2020	95.1%	4.6%	0.3%
Rifamycin/isoniazid (active and latent TB, other infections)	2021	97.7%	2.1%	0.2%
Rifamycin/isoniazid (active and latent TB, other infections)	2022	99.0%	0.8%	0.2%
Rifamycin/isoniazid (active and latent TB, other infections)	2023	99.0%	0.8%	0.2%
Rifamycin/isoniazid (active and latent TB, other infections)	2024	99.2%	0.6%	0.1%
IL-6 inhibitors and abatacept	2019	90.1%	8.7%	1.2%
IL-6 inhibitors and abatacept	2020	93.9%	4.9%	1.2%
IL-6 inhibitors and abatacept	2021	96.3%	2.8%	0.9%
IL-6 inhibitors and abatacept	2022	97.8%	1.0%	1.2%

Drug group	Year	Direct	Successor	Unmatched
IL-6 inhibitors and abatacept	2023	97.9%	0.5%	1.5%
IL-6 inhibitors and abatacept	2024	97.8%	0.4%	1.8%
JAK inhibitors (rheumatology)	2019	90.1%	8.9%	0.9%
JAK inhibitors (rheumatology)	2020	94.8%	4.4%	0.9%
JAK inhibitors (rheumatology)	2021	97.1%	2.5%	0.4%
JAK inhibitors (rheumatology)	2022	98.5%	1.1%	0.4%
JAK inhibitors (rheumatology)	2023	98.9%	0.4%	0.7%
JAK inhibitors (rheumatology)	2024	98.4%	0.3%	1.3%
Low-TB-risk biologics (negative control)	2019	91.9%	7.9%	0.2%
Low-TB-risk biologics (negative control)	2020	96.0%	3.9%	0.1%
Low-TB-risk biologics (negative control)	2021	97.9%	2.0%	0.1%
Low-TB-risk biologics (negative control)	2022	99.1%	0.8%	0.1%
Low-TB-risk biologics (negative control)	2023	99.5%	0.4%	0.1%
Low-TB-risk biologics (negative control)	2024	99.6%	0.2%	0.2%
Levetiracetam (negative control)	2019	87.6%	7.5%	4.9%
Levetiracetam (negative control)	2020	91.3%	4.2%	4.5%
Levetiracetam (negative control)	2021	93.5%	2.3%	4.2%
Levetiracetam (negative control)	2022	94.7%	1.2%	4.0%

Drug group	Year	Direct	Successor	Unmatched
Levetiracetam (negative control)	2023	94.8%	0.9%	4.3%
Levetiracetam (negative control)	2024	95.0%	0.7%	4.3%
Rituximab	2019	91.2%	8.4%	0.4%
Rituximab	2020	95.1%	4.4%	0.5%
Rituximab	2021	97.5%	2.1%	0.5%
Rituximab	2022	98.7%	0.8%	0.5%
Rituximab	2023	99.0%	0.4%	0.5%
Rituximab	2024	99.2%	0.2%	0.6%
Systemic glucocorticoids	2019	90.6%	8.8%	0.5%
Systemic glucocorticoids	2020	94.6%	4.8%	0.6%
Systemic glucocorticoids	2021	97.3%	2.2%	0.5%
Systemic glucocorticoids	2022	98.4%	1.1%	0.5%
Systemic glucocorticoids	2023	98.8%	0.7%	0.5%
Systemic glucocorticoids	2024	99.0%	0.5%	0.5%
TNF inhibitors	2019	91.2%	8.2%	0.6%
TNF inhibitors	2020	95.3%	4.1%	0.6%
TNF inhibitors	2021	97.6%	2.0%	0.4%
TNF inhibitors	2022	98.5%	0.9%	0.6%
TNF inhibitors	2023	98.6%	0.4%	0.9%
TNF inhibitors	2024	98.9%	0.3%	0.9%
Calcineurin/mTOR inhibitors	2019	92.3%	7.6%	0.1%
Calcineurin/mTOR inhibitors	2020	96.3%	3.6%	0.1%
Calcineurin/mTOR inhibitors	2021	99.3%	0.6%	0.1%
Calcineurin/mTOR inhibitors	2022	99.8%	0.1%	0.1%
Calcineurin/mTOR inhibitors	2023	99.9%	0.0%	0.1%
Calcineurin/mTOR inhibitors	2024	99.9%	0.0%	0.1%